



MATHS in Event Management



Barbara Lynch and
Angelo Pantalone



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resources



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Angelo Pantalone

This book was commissioned by Jennifer Gough, project management, editing and typesetting was by Render Editorial Services, and Gail McManus Graphics designed the cover.

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Printed in Australia by Ligare Book Printer

Blake
EDUCATION
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Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au



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ISBN 978 1 92571 018 2



Maths in Events Management is a collection of photocopiable student work sheets which feature the application of mathematical knowledge and techniques to the planning, organising, managing and marketing of a range of events, from parties, dinners and weddings to art exhibitions and music festivals. Any information or theory required in a subject area is provided in the work sheet. Students are given an 'idea' of the answer in some cases.

Event management is becoming an increasingly popular profession because it allows people to be involved in a range of projects, to meet celebrities and to attend exciting events. People are attracted by the range of skills that they will learn, such as marketing, communication, problem-solving, time and project management and, of course, dealing with people.

The series has been designed to motivate and interest students of Mathematics in Secondary Schools, particularly at Year 9 or 10 level, and is suitable for students who require additional Mathematics for TAFE courses in Event Management. In most cases, an electronic calculator or spreadsheet program would be useful.

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History is full of interesting information about events. Events can be social functions, dinners, sporting events, award nights, fetes, launches and 'thons', such as walkathons. In Australia, special events are often used to promote travel and tourism too.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 As more companies developed events over the last 5 years in Australia, the amounts raised after costs showed an 85% growth to \$2.8 million. How much in whole number dollars was raised through events 5 years ago?

Idea: $x \times 1.85 = 2.8$ million

Special events

- 2 It was suggested that the full Greek national anthem be sung at the opening of the Olympic Games in Athens in 2004. How many verses are said to be in the Greek national anthem?

Idea: (the area of a square of side length 12 units) – 4

- 3 It was estimated that 500 000 people attended the 1969 Woodstock Music Festival, held on a 2 400 000 m² farm in the United States. Six hundred toilets were provided.

- a How many square metres per person were available?
b How many people per toilet were there? Give your answer as a whole number.

- 4 The opening of the Sydney Harbour Bridge was a grand event with fireworks, a parade of floats and marching bands.

- a How many years did it take to build the bridge?

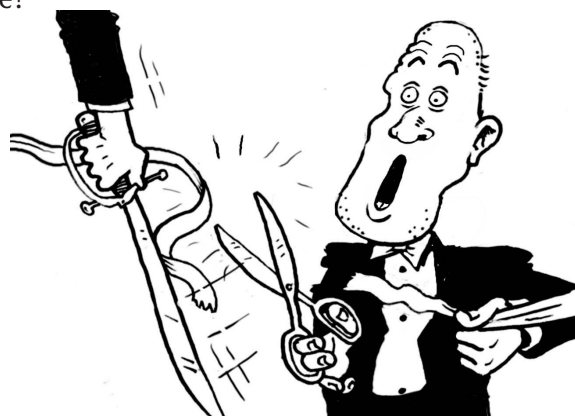
Idea: (the area of a triangle with base length 2 units and height 8 units)

- b How many people watched the opening, when a horseman cut the ribbon ahead of Premier Lang?

Idea: $36 \text{ million} \div 480$

- c In what year was the opening?

Idea: $2 \times 3^2 \times 10^2 + 11 \times 2^2 \times 3$



Sporting events

- 5 Approximately how many Australians participate in organised sport? Give your answer to the nearest million.

Idea: 27% of the population of Australia (22 043 453)

- 6 Netball is the most popular sport for females in Australia. How many Australian girls and women play netball every month?

Idea: $182\,535 + 77\,465 - 20\,000$

- 7 Many international hockey tournaments are hosted in Australia. For how many years has hockey been played?

Idea: $\sqrt{90\,000}$

... FAST MATHS FACTS

Area of a triangle =
 $\frac{1}{2} \times \text{base length} \times \text{height}$

Area of a square =
(side length)²

Speed (ms⁻¹) = $\frac{\text{distance (m)}}{\text{time (s)}}$

Order of operations:
BODMAS

1 kilometre = 1000 metres

A prime number has only two different factors: itself and one.



- 8** Professional bike races such as the Tour de France are major sporting events. How old was the youngest winner of this bike race?
Idea: (the largest prime number between 2 and 20)
- 9** When did the first official women's boxing match take place? *Idea:* $7 \times [(75 \times 3) + 2 \times (17 - 5)] + 133$
- 10** Golf tournaments are major events in which the prizes can be hundreds of thousands of dollars.
- a** In which year did the Scottish Government ban a golf tournament? *Idea:* 61×27
- b** How long, in metres, is the longest hole on any golf course in the world?
Idea: $16 \times 49.149 \text{ m}$
- c** A golf ball travelled from the first tee to the first green, a distance of 300 m, in $15\frac{1}{2}$ seconds. What was the average speed of the golf ball in metres per second?
- 11** The World Cup is played on a soccer field that is 100 m long. In an average game, a linesman runs the length of the field 64 times. How many kilometres does he run altogether during a game?
- 12** In a cricket test match, what is the minimum number of times a ball may be bowled before it is replaced?
Idea: finish the number pattern $(3 \times 2) \times (4 \times 2) \times (\dots \times 2)$ and calculate

... FAST EVENT FACTS

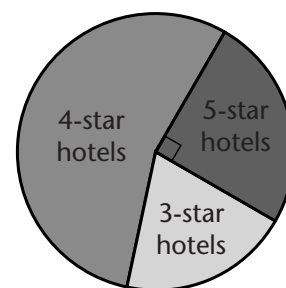
A **special event** is a function or occasion that aims to raise money, raise awareness of an issue, get publicity or improve a group's reputation.

Events can be large or small, simple or complicated.

The **key to a successful event** is good planning and good management.

Tourism and events

- 13** What is the area in m^2 of the huge hotel that is carved out of ice in a village in Sweden?
Idea: $103 \times 22 \text{ m}^2$
- 14** How many hundred million jobs are there in the travel and tourism industry worldwide?
Idea: $\frac{15.75 + 9.312}{3} - 1.72$
- 15** The pie chart, right, shows the proportions of different standards of hotels in Australia. The more stars a hotel is awarded the better it is rated. In a particular city there are 6428 hotels, with an average of 96 rooms each.
- a** How many hotel rooms are available in that city?
- b** According to the pie chart, how many hotels are rated as 5-star?
- 16** In 2012, a newspaper reported, 'There were one third fewer marquees at this year's Melbourne Cup carnival than two years ago'. In 2012, there were 34 corporate tents, compared with 53 in 2010. Calculate the percentage fewer and hence state whether the headline was correct. Give your answer correct to 2 decimal places.
- 17** The Mayor of Podsville said: 'The 91 000 visitors to the week-long Food, Wine and Lifestyle event are expected to bring more than \$10 million into the city's local economy'. If this is true, how much is each visitor expected to spend?



Many events involve paying big money to movie stars and other celebrities to help promote the occasion. These stars can influence the success of the event. This is particularly true of big events in areas such as sport and entertainment. Big numbers have lots of numbers and zeros, and you have to be careful that you don't make silly mistakes.

- 1 Using 1s and 0s, write the answer to 1 million + 1 billion + 1 trillion + 1 quintillion + 1 sextillion + 1 nonillion.
- 2 A googol is a 1 with a hundred zeros. Write a googol as a power of 10.
- 3 A googolplex is the biggest number with a name. A googolplex is 1 with googol zeros. Write a googolplex as powers of 10.

Money, money, money

- 4 Convention centres are used for events such as trade fairs and conferences. In 2005 in the United States, \$2.4 billion was spent building 44 new centres. How much did each centre cost on average? Give your answer correct to the nearest whole number.
- 5 A state events development program has invested \$9.7 million in 460 events since it began in 2001. How much did it spend on each event on average? Give your answer correct to the nearest whole number.
- 6 When event organisers invited the golf champion Panther Forrest to come to Australia to play in a local tournament, one of the state governments agreed to pay half of his \$4.5 million appearance fee.
 - a How many million dollars did the government pay Panther?
 - b The government stated that Panther's appearance would generate \$19 million in economic benefits. What is the government's payment as a percentage of the estimated economic benefits? Give your answer correct to 2 decimal places.
 - c A sample of 500 people was asked whether the state government should contribute to Panther's fee: 65% said 'no'. How many people said 'no' to the question?
- 7 Fred, a football official, is paid \$1.58 million each year. Surveyed football fans said that he should be paid just $\frac{3}{4}$ of that figure. How much did they think he should be paid?
- 8 The Major League baseball competition has a very high profile and sponsors are keen to become involved. Peter Pitchen, a star player, recently signed a contract worth \$250 million. There are 1200 players in the competition. If every player in the competition was to receive the same amount as Peter, how much would this cost altogether? What is this amount in trillion dollars?

... FAST NUMBER FACTS

1 (one)
 10 (ten)
 100 (one hundred)
 1000 (one thousand)
 10 000 (ten thousand)
 100 000 (one hundred thousand)
 1 000 000 (one million)
 10 000 000 (ten million)
 100 000 000 (one hundred million)
 1 000 000 000 (one billion)
 10 000 000 000 (ten billion)
 100 000 000 000 (one hundred billion)
 1 000 000 000 000 (one trillion)
 1 followed by 18 zeros (one quintillion)
 1 followed by 21 zeros (one sextillion)
 1 followed by 30 zeros (one nonillion)
 1 followed by 303 zeros (one centillion)

Time and money

... FAST TIME FACTS

A million seconds is 12 days.

A billion seconds is 31 years.

A trillion seconds is 31 688 years.

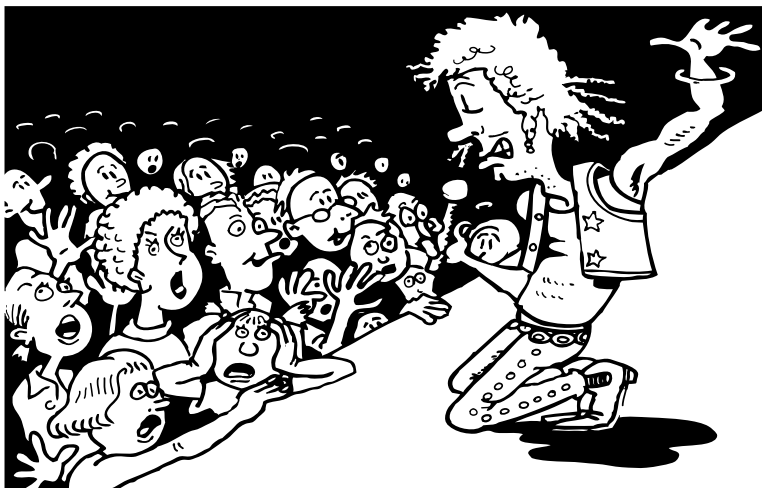
- 9 Velvet Rock will perform a concert if they are paid \$1 for every second they are on stage.

a Complete the following calculations to find how much Velvet Rock would receive if they worked a 40-hour week for all 52 weeks of the year:

- 1 minute (min) = 60 seconds (s)
- 1 hour (hr) = 60 min
- 40 hr = 40 × 60 min =
40 × 60 min × 60 s =
..... s
- 52 weeks = 52 × s =
..... s
- Amount earned in dollars =
\$

b How much would Velvet Rock earn if they were paid for every second of a 365-day year?

c How many years would they need to work if they charged \$1000 per second, worked every second of the year and wanted to earn \$1 trillion? Give your answer to the nearest year.



Charity fundraising and events

Charities organise events that raise money for good causes. Charitable foundations are good sources of funding for the Arts and also in situations where disaster relief is needed.

- 10 When was the New South Wales Society for Promoting Christian Knowledge and Benevolence, one of the oldest charitable foundations in Australia, established?

Idea:
$$\frac{2.02 \times 3^2 \times 120 \times 10^6}{2^2 \times 3 \times 10^5}$$

- 11 In one year, 100 leading corporations in Australia contributed \$220 million and over 200 000 staff hours to community events and \$290 million and 425 000 staff hours to environmental projects. How much money and how many hours did the leading corporations contribute?
- 12 After the Indian Ocean tsunami in 2004, 13.4 million Australians organised events that raised a total of \$7.7 billion to assist the tsunami victims.
- a How much, on average, did each Australian raise?
 - b Of the \$7.7 billion, \$5.7 billion was given in donations and \$2 billion was raised through charity events and 'charity gambling', such as raffles. What percentage of the total funds was raised through charity events? Give your answer correct to 2 decimal places.
- 13 Between 2003 and 2004, 67% of 784 925 businesses gave a total of \$3.2 billion to charities.
- a How many businesses gave to charities? Give your answer to the nearest whole number.
 - b Of the 67% of businesses that donated, how much on average did each business give to charities? Give your answer to the nearest dollar.

Jan runs Great Occasions, an event management company that organises everything from birthday parties to car races. Jan has been so busy that she has decided to hire Lilli to help her. Lilli has no experience in the event management business so Jan has made a work sheet that will give Lilli all the information she needs to know!

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

Bread

- 1 In a 400 g loaf of wholemeal bread, each slice weighs 33 g. Approximately how many whole slices are there in a loaf?
- 2 Low-kilojoule thin-sliced bread is sold in 400 g loaves. If each slice weighs 20.6 g, approximately how many whole slices are in a loaf?
- 3 a How many slices in a standard loaf of bread?

Idea: $\frac{2^5 \times 3^2}{12}$

- b The small sandwiches for a cocktail party are made of three slices of bread, each separated by a different filling. The sandwich is then cut into quarters. How many of these sandwich quarters can be made from a standard loaf of bread?



Appetisers

- 4 How many appetisers per person are enough for a 1.5 hour cocktail party?

Idea: between $\frac{10^3 \times 10^2 \times 10^4}{10^5 \times 10^3}$ and $\frac{2^5 \times 3^4}{2^3 \times 3^3}$

- 5 How many appetisers per person are enough for a dinner party?

Idea: $\frac{2^4 \times 3^4}{2^3 \times 3^3}$

- 6 a Jan explains to Lilli that to calculate the number of appetisers needed for a cocktail party she should use the following formula:

$$\text{Number of appetisers} = 12 \times \frac{\text{number of people}}{\text{number of different sorts of appetisers}}$$

Use the information in the Fast Appetiser Facts table to find how many appetisers are needed for a cocktail party for 25 guests.

- b If serving appetisers before a dinner party, half the number found by this formula should be served. Use the Fast Appetiser Facts information to find how many appetisers are needed before a dinner party for 8 people.

... FAST APPETISER FACTS

Number of guests	Number of different types of appetisers
8–10	3
14–16	4
45 or less	6
More than 45	8

Half the appetisers should be served hot and half cold. If this is an odd number, have more cold than hot.

- 7** For a cocktail party, each guest needs 5 pieces of food for the first hour and then 4 pieces of food per hour after that.
- How many pieces of food will be needed per person for a cocktail party that starts at 7 pm and finishes at 10 pm?
 - How long was a party if each person had on average 17 pieces of food?

Space needs

- 8** A tent measuring 9 metres by 12 metres will hold 100 guests.
- What area in m^2 is needed for 100 guests?
 - What area in m^2 is needed for each guest?
 - You will also need extra room in the tent for buffet tables and the caterer's equipment. If a buffet table is 1.8 m long and 1 m wide, how much space will it take up in the tent?
 - You need one table for every 75 guests. What area is taken up by tables?

Wait staff

- 9** The number of wait staff needed for a cocktail party is 1 for every 20 guests. How many wait staff will be needed for a party of 95 guests?
- 10** The number of wait staff for a sit-down dinner is 1 for every 8 guests.
- How many wait staff will be needed for a party of 95 guests?
 - How many wait staff are needed altogether if a butler is needed to supervise the wait staff?

Drinks

- 11** How many glasses of wine are there in a 750 mL bottle if a glass contains 120 mL? What about if it contains 180 mL, the standard pour at the bar? Give your answers in fraction form.

... FAST SPACE FACTS

For a cocktail party at which all the guests will be standing up, you need 0.5 m^2 of space per guest.

For a cocktail party at which some guests will be standing and some sitting, you need 0.75 m^2 per guest.

For a dinner party using round tables for 6, 8 and 12, you need 1.25 m^2 per guest.

For dancing, you need 0.2 m^2 to 0.5 m^2 per guest.

- 12 a** How many 225 mL flute glasses can be filled to the brim with champagne from a 750 mL bottle? Give your answer in fraction form.
- b** Because champagne fizzes, you should only fill a flute to $\frac{3}{4}$. How much champagne should you put in a 225 mL flute?
- c** Considering this, how many flute glasses can you really pour from one standard bottle of champagne?
- 13** A wedding is always a great celebration. A wedding reception usually runs for the whole evening, over five or six hours. Jan caters for 4 drinks per guest for the night. How many drinks will she need for 150 guests?
- 14** Guests at a cocktail party will often forget where they left their glass, therefore you will need twice as many clean glasses as guests. How many glasses will be needed for a party of 95 guests?
- 15** You need ice to keep the drinks cold. You will need 5 kg of ice for every 10 guests.
- What mass of ice is needed for each guest?
 - If each ice cube weighs 14 grams, approximately how many ice cubes will each guest receive?
 - How many ice cubes will be needed for 150 guests? Give your answer to the nearest hundred.

There are risks in managing an event: risks to human life, the environment, the organisation hosting the event, the operation or project, and to the reputations of the organisation and other people. It is important to insure against some eventualities and to prepare for others.

Be insured!

- 1 At least 5% of the entire budget for an event should be allowed for unforeseen happenings. If the entire budget for a street fair is \$25 000, how much should be allowed for emergencies?
- 2
 - a An insurance company earned \$4 million in premiums and paid out only \$200 000 to victims of accidents. What percentage of the premiums was paid to victims?
 - b How much profit did the insurance company make?
- 3 A school fete had 15 market stalls and the organisers needed to insure each stall separately.
 - a The organisers must take out public liability insurance to cover any injuries to people who attend the fete. The minimum insurance premium for each stall is \$255 for \$10 million cover. How much will it cost for all the stalls?
 - b The organisers must also take out product liability insurance to cover any injury caused by what they are selling. The minimum insurance premium for each stall is \$110 for \$10 million cover. How much will this cost for all the stalls?
 - c How much will the insurance cost altogether?
- 4 Entertainers such as musicians, clowns, puppeteers and face-painters must be insured. For \$10 million cover, the premiums vary as shown in the table, right.
 - a Action Entertainment will provide a fair with 2 solo singers, 1 face-painter, a magician double act and a 10-person puppet group. How much will insurance cost?
 - b If the performers are required to share the insurance cost equally, how much must each pay?
- 5 Between 2010 and 2012, the average claim on events public liability insurance at Ecco Insurance Company (EIC) increased from \$55 000 to \$88 000.
 - a By what percentage did the average claim change between 2010 and 2012?
 - b EIC wants to put up its premiums in 2013, using the same annual rate of change as the average claim over 2010–12. If the average premium in 2012 was \$225, what should be the premium in 2013?

... FAST RISK FACTS

Insurance is used to protect you from loss of money due to unforeseen incidents or accidents.

An **insurance policy** is a contract in which an insurance company agrees to pay for any losses covered by the policy.

The **insurance cover** is the maximum amount that will be paid by the insurance company.

An **insurance premium** is what you pay to an insurance company for the policy.

Public liability insurance is a policy that covers death or injury to anyone at an event who is not an employee.

Product liability insurance is a policy that covers you for any injury or damage caused by a product you supply.

Act	Premium
Solo performer	\$220
Double act	\$300
3–5 person group	\$500
6–8 person group	\$715
9–12 person group	\$880
13–15 person group	\$1 070

Promising weather

The weather can be the difference between an event's success or failure. Forecasting the chance of rain is very difficult, since the probability of rain usually changes from day to day.

6 The probability that it will rain during Saturday's walkathon is said to be 20%. What is the probability that the weather will be fine?

7 Philip's Event Company is running an outdoor art show from Monday to Friday, inclusive. If it rains, the show will be ruined. Philip asks a weather forecaster the probability that it will rain some time during Monday, Tuesday, Wednesday, Thursday or Friday. He is told that at this time of the year every day has a 15% probability of rain.

- What is the probability that it won't rain on a particular day? Call this q and express it as a percentage.
- The way to work out the probability of no rain during the show is to calculate:

$$100\% - q \times q \times q \times q \times q = 100\% - \dots\% = \dots\%$$

Give your answer correct to 1 decimal place.



... FAST PROBABILITY FACTS

Long-run proportion or relative frequency after a very large number of trials =

$$\frac{\text{number of favourable outcomes}}{\text{total number of trials}}$$

The **probability** of a random occurrence is the relative frequency of the occurrence if the trials were repeated over a very long time.

The **probability of something not happening** = $1 - \text{probability of it happening}$

We can't afford it!

Sponsors are interested in how much they can earn from their investment in your event. You need to prove that you will be able to give them effective advertising of their product. However, it can be a risk to rely on sponsorship alone to run an event. Promises of sponsorship may not be kept!

8 For the Boomers Expo, sponsorship is available at three levels: gold, costing \$30 000; silver, costing \$10 000; bronze, costing \$5 000. Only 4 bronze, 4 silver and 1 gold sponsorships are available.

- How much will the organisers receive if all the sponsorships are taken up?
- How much will organisers lose if the gold and 2 of the silver sponsorship promises are not kept?

... FAST SPONSORSHIP FACTS

A sponsor is a person or company who provides products, services or money in exchange for special benefits such as advertising, a list of attendees or free attendance.



Alex and Ellie have written books on the importance of Mathematics in a number of occupations. Their latest book is about event management and they decide that they should manage an event to launch the book. They have a small budget so they need to keep costs to a minimum. They have plenty of time and will do much of the work themselves.

What about the date?

- 1 A book takes a long time from when it is first written until it arrives at the warehouse, ready to go to the shops. A book launch will be booked the day the manuscript is delivered to the publisher. The launch will be held the day that the books arrive at the warehouse. According to the schedule, right, how many days will it take to produce the book, excluding the holiday break?
- 2 They have decided to have a private party for the launch and invite 50 people. This means that there will be no advertising costs. Alex has designed the invitations and Ellie will print them on colourful cards. They bought the 50 cards for 65 cents each and paid 35 cents for each matching envelope. They will address the envelopes themselves but will need to pay 55 cents for each stamp. How much will the invitations cost altogether?

Action	Dates
Manuscript delivered to publisher	1 November 2014
Publisher sends manuscript to reviewer	1–30 November 2014
The book is rewritten where necessary	1–31 December 2014
Holiday break	1–31 January 2015
Final manuscript delivered for final review	1–7 February 2015
Managing editor finalised files	8–14 February 2015
Book in production	15 February – 14 May 2015
Book is printed	15–31 May 2015
Printed books arrive at warehouse	1 June 2015

Where will we go?

Restaurant or bar	Number of people for cocktail party	Number of people seated	Minimum spend*
Riley’s Restaurant	not available	34	\$2 000
Rooftop	40	not available	\$1 500
Spanish Lounge	50	not available	\$3 000
Zach’s Lounge	30	20	\$2 000
Formal Dining Room and Restaurant	not available	70	\$4 000
*The minimum spend is the smallest amount a venue charges so that it can meet all its expenses.			

- 3 A big hotel in the city has a number of rooms that could be used for the launch. The authors think that a cocktail party will be cheapest.
 - a Alex and Ellie are concerned about the ‘minimum spend’ because they will have to spend this amount on food and drink no matter how many people come. What percentage of the 50 people invited need to accept so that the authors can book a venue with a minimum spend of:
 - i \$1500
 - ii \$3000?
 - b Alex and Ellie decide they will be able to use Zach’s Lounge for the event. What fraction of people invited accepted the invitation?

Booking and party packages

- 4 a A minimum deposit of \$500 is required when the booking is confirmed. An extra 15% must be paid if the party has more than 10 people (which the book launch has). All this money will be deducted from the final account. How much needs to be paid up-front?
- b Final numbers are required 7 days prior to the function date. This is the number that will be catered for and paid for. On what date will the final numbers need to be given to Zach's Lounge?
- 5 The prices for party packages by Zach's Lounge are set out in the table, right.
- a Zach's Lounge has a minimum spend of \$2000. Alex and Ellie can afford \$300 more than that. Which of the following alternatives can they choose from if 30 people will be attending?
- Premium package with champagne and cocktail for 2 hours
 - Standard package with cocktail for 4 hours
 - Premium package with no extras for 3 hours
 - Premium package with champagne for 3 hours
 - Standard package with no extras for 4 hours
- b Full payment for the event must be made on or before the day of the event. If payment is by credit card, an extra fee of 3% is charged on the amount that is over \$1000. If they choose a \$2280 package, how much will be saved by not paying by credit card?

Length of time	Standard package, price per person	Premium package, price per person
2 hours	\$40	\$56
3 hours	\$60	\$76
4 hours	\$80	\$96
5 hours	\$100	\$116

- Champagne for toasts: add \$2 per person to package
- Cocktail on arrival: add \$10 per person to package

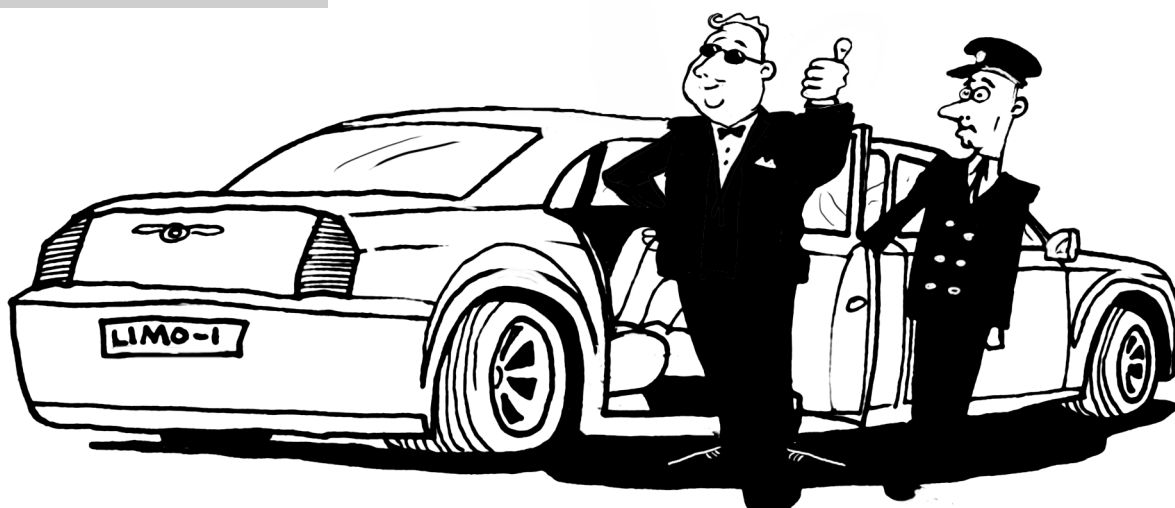
Launch pad

... FAST LIMO FACTS

Hire costs for a Rolls Royce Phantom limousine:

- 2 hours: \$550
- 3 hours: \$650
- 4 hours: \$750
- 5 hours: \$800

- 6 Famous events promoter Eddie O'Dwyer has agreed to launch the book for free – but he needs to be provided with a Rolls Royce limousine to and from the venue, as well as free cocktails and canapés! If the chosen party package is \$58 per head, use Fast Limo Facts to find how long (including travel time) Alex and Ellie can afford to have Eddie attend if they want to spend less than \$800 on his expenses.



A quiz show needs to be organised like any other event. The manager needs to work out the theme of the quiz, how many contestants will participate, who will be in the audience, how long the show will last and what the prizes will be.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

How long is the show?

- 1 The show, including commercials, will take 30 minutes. Commercials take 5% of this time. How much time would be left for the quiz?
- 2 There will be 6 contestants. How much time on average will each contestant have to answer questions?
- 3 Fifteen seconds are allocated for each question. How many questions, on average, would each contestant have to answer?

What will the contestants win?

- 4 Each of the 6 contestants receives \$100 appearance money. After the first round of questions, the 2 contestants who got the most questions wrong are eliminated, leaving 4 contestants. Each of them receives \$200. After the second round of questions, the 2 contestants who got the most questions wrong are eliminated, leaving 2 contestants in the grand final. Each of them receives \$400. The winner of the grand final receives \$1000.
 - a How much is needed altogether to pay for the prizes?
 - b How much will the winner have received at the end?



How many people will be in the audience?

- 5 The auditorium holds 100 people and this evening there are 40 men and 60 women in the audience. It is decided to have 6 contestants selected from the audience.
 - a What percentage of people in the auditorium are women?
 - b The sponsor of the quiz is Kleen washing powder. They want the first contestant on the show to be a woman. What is the chance that this will be the case?
 - c What is the chance that all 6 contestants chosen will be women? Give your answer correct to 3 decimal places.

Idea: $\frac{60 \times 55 \times 58 \times 59 \times 56 \times 57}{95 \times 99 \times 100 \times 97 \times 96 \times 98}$

- d What is the chance that half of the 6 contestants chosen will be women? Give your answer correct to 3 decimal places.

Idea: $\frac{60 \times 20 \times 38 \times 40 \times 39 \times 58 \times 59}{99 \times 95 \times 98 \times 100 \times 96 \times 97}$

Show me the money! (cont'd)



6

Okay! Hands on buzzers!

6 For the grand final, contestants select questions from the grid below. The question cards given to the host must have the answers on them as well. Complete these cards for the host.

a Spaced out!

How much space does a rectangular prism take up if it measures 2 cm by 3 cm by 4 cm?

Answer: cm³

b Unknown quantity

A person thinks of a number, multiplies it by 4 and adds 5. If the answer is 25, what was the original number?

Answer:

c Chances are

What are the chances that if a contestant selects a question from this grid without looking, it will be about money?

Answer:

d Pie in the sky

An old-fashioned recipe for apple pie asks for 4.4 pounds of apples. How many kilograms are needed if there are 0.454 kg in a pound?

Answer: kg

e Bookworm

The page of a book measures 160 mm by 110 mm. A bookworm ate 5 pages. How many square centimetres of paper did the bookworm eat?

Answer: cm²

f Model plane

The model plane was built to scale with 1 cm representing 1 m. How long are the wings on the model if the actual plane's wings are 50 m long?

Answer: cm

g Evergreen trees

The tree grew to be 5 times as high in 2 years. If it continues to grow at the same rate, how tall will it be compared with its original height in another year's time?

Answer:

h Cordially yours

Lemon drink is made with cordial and water in the ratio of 2:7. How much water is needed to make 900 mL of lemon drink?

Answer: mL

i Sale! Sale! Sale!

A mobile phone is on sale for \$540. If this sale price is 10% cheaper than the original price, how many dollars cheaper is it?

Answer: \$

Winner takes all!

7 The sponsor decides they want part of the competition to be between two teams, each of 3 contestants. If a contestant answers a question correctly, the team wins 2 points. If a contestant answers a question incorrectly, the team loses 1 point.

The results of the competition are in the table, right. Which team won and with how many points?

Team Surfer			Team Skater		
Question	Correct?	Points	Question	Correct?	Points
1	✓		1	✗	
2	✗		2	✓	
3	✓		3	✗	
4	✓		4	✗	
5	✗		5	✗	
6	✓		6	✓	
7	✓		7	✓	
8	✗		8	✓	
9	✓		9	✗	
10	✗		10	✓	
Total			Total		



With some of his Year 11 school friends, Jackson is organising a fundraising event in support of the homeless. It will be a walkathon in which students and guests are paid for each kilometre walked. He decides to call his fundraising event Homerun!

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

Okay, how many joggers can we expect?

- 1 There are 190 students in Year 11 at Jackson's school. The year level below has 10% fewer students, while the year level above has 20% more students than Jackson's year. Project Homerun is a 12-km walk.

a Calculate the total number of students in Jackson's school in Years 10, 11 and 12.

b If an additional 65% of this number of students are expected to take part in the event, calculate the likely number of walkers.

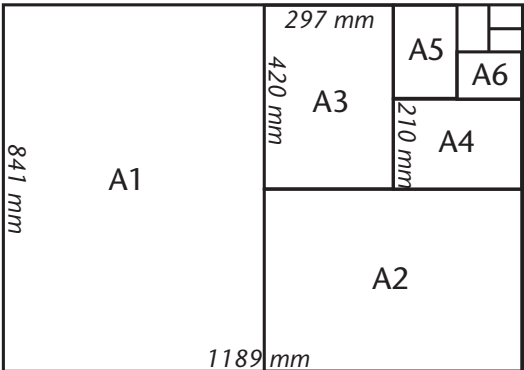
c Using the Fast Jogging Fact, convert the 12-km walkathon into miles, correct to the nearest whole number.

d If the number of students calculated in Question 1b actually walks, what will be the total distance in kilometres walked by all the participants?
- ... **FAST JOGGING FACTS**

1 kilometre = 0.621371192 miles

A bit about paper

- 2 Paper comes in different sizes, and the most common sizes belong to the A series. A4 size is the one that is usually used for letters and photocopy paper. An A4 sheet of paper folded in half gives two A5 sheets, and an A5 sheet folded in half gives two A6 sheets. How many A6 sheets can be made from an A4 sheet of paper?



Put your best foot forward

Before he kicks off the Homerun event and starts promoting it, Jackson must calculate expenses against the amount of money he will receive.

- 3 a Calculate the dollar values of each of the five activities. Write them in the 'Cost' column of the table.

b Calculate the total expenses for the event.

c Calculate the time spent managing the five activities.

No.	Item	Time spent	Detail	Cost
1	Computer-designed A6 flyer with high-grade colour printout	3 hours	\$15.00 (\$5.00 per hour) + \$2.50	
2	Printing of 600 A6 flyers	2.5 hours	60 cents for one A4 copy	
3	Distribute flyers to students in the 3 school year levels	1.5 hours per year level	Free	
4	Inform both town halls (2 calls and 2 letters)	30 minutes	50 cents + two 55 cent stamps	
5	Establish a contact who handles the money	1 hour	Free – using PE teacher, Mrs Costalot	

Show me the money!

The 12-km walk starts at the Melbourne Town Hall, in the central business district (CBD), and ends at the Prahran Town Hall. Aqua Inc. has donated the water that will be consumed during the Homerun event.

- 4 a Can you remember the expected number of walkers?

Idea: $\frac{5832}{4} - 486$

- b If each walker consumed 250 mL of water, what would be the entire volume consumed in litres?
- c How much money could be raised if each walker paid \$3 per kilometre?
- d What would be the net money raised after expenses for the event? (Use the answer to Question 3b.)

That's some storm!

The weather bureau forecasts a big storm that will coincide with the walkathon, and a number of walkers decide to drop out!

- 5 a The expected number of 972 walkers has been reduced by 21.5% due to the weather conditions. How many are now participating in the event?
- b How many kilometres in total will be walked by the reduced number of walkers?
- c At \$3 per kilometre, what is the expected money for the reduced numbers?
- d What would be the net money raised after expenses for the event with the reduced numbers? (Use the answer to Question 3b.)
- 6 The charity HomeFind is thankful to the Homerun organisers and the school. HomeFind calculates that 65 homeless people can be accommodated for a month using the money raised, even though there were reduced numbers of walkers.
- a Calculate the rounded dollar amount required to place each of the 65 people into a house. (Use your answer to Question 5d.)
- b Of the 65 houses, 10 have room for two people rather than one. How many people can then be placed?
- c Calculate the rounded dollar amount required to place each of these people into a house.
- d At the paying rate of \$3 a kilometre, how many kilometres need now be walked to place one person into a house? Give your answer correct to 1 decimal place.





Frances is the principal of Fairview High School. She has been at the school for a very long time and will retire soon. All the parents want to give her a farewell party and there is a great deal of discussion about what kind of party to have.

1 Frances leaves on the afternoon of 21 December. It is already 1 October, so the parents consult the calendar.

October							November							December						
Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa	Su	Mo	Tu	We	Th	Fr	Sa
	1	2	3	4	5	6					1	2	3							1
7	8	9	10	11	12	13	4	5	6	7	8	9	10	2	3	4	5	6	7	8
14	15	16	17	18	19	20	11	12	13	14	15	16	17	9	10	11	12	13	14	15
21	22	23	24	25	26	27	18	19	20	21	22	23	24	16	17	18	19	20	21	22
28	29	30	31				25	26	27	28	29	30		23	24	25	26	27	28	29
														30	31					

- a

The parents have decided that the best nights for the party would be Friday or Saturday night. How many choices do they have for the party if they do not wish to hold it on Frances’ last night?
- b

If they need at least 5 weeks to plan the party, what is the earliest date the party could be held?
- c

If they hold the party on the last Saturday before Frances leaves, how many days will the parents have to prepare?
- 2

The school will allow a party to be held in its large assembly hall, with 450 people for 3 hours. It has the advantage that there would be no venue charge. The music would also be free because the school band can play, and the parents only need to provide drinks and food. There would be a charge of \$8 per hour per person for drinks. How much, in total, would it cost for drinks?
- 3

One of the parents has offered to make platters of canapés, the small appetising ‘finger foods’ that are easy to eat standing up. Each canapé will cost \$2 and each guest is expected to eat 5 canapés per hour during the 3-hour party. How much would this cost:

a

for each of the 450 guests?

b

for all the canapés?
- 4

The parents have two catering quotes for food and drink for 450 people for 3 hours:

• Jan’s Affairs: \$21 per person for 6 appetisers each, and \$35 per person for drinks for the 3 hours

• Easy Events: \$50 each for platters for 10 people each with 4 dips, and \$10 per hour per person for drinks.

a

How much will it cost for all the guests if Jan’s Affairs is catering?

b

How much should be charged for each guest if Jan’s Affairs is catering?

c

How much will it cost for all the guests if Easy Events is catering?

d

How much should be charged for each guest if Easy Events is catering?

e

Which catering will be the cheapest for the guests?
- 18

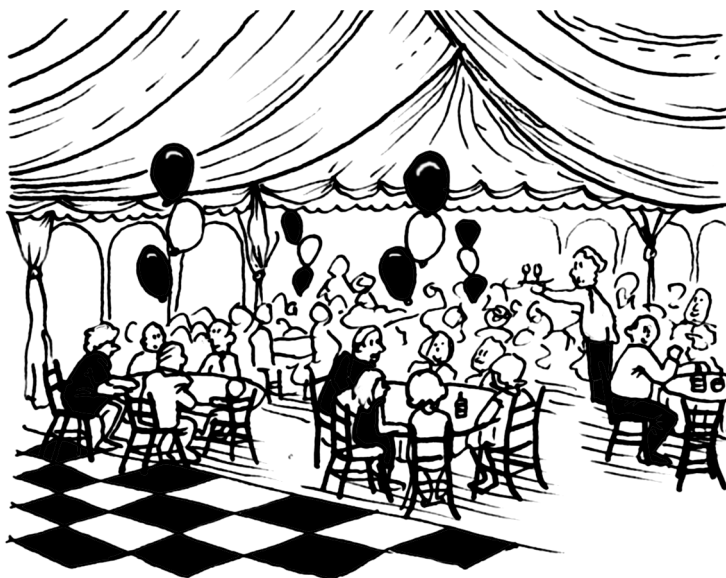
Maths in Event Management

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Maths in Event Management

- 5** Usually the food and drink cost is between 35% and 50% of the cost of the whole party if wait staff costs are not included. The farewell party for the previous principal cost \$40 000. What was the range of the cost of the food and drink?
- 6** Frances would prefer a more stylish party. This means hiring a venue, wait staff and musicians, and having food and drinks provided by the venue. Frances believes the perfect venue is La Courgette. It has a space to host a cocktail party for 320 people. Since it is clear that the cost will be higher than if the school hall were used, the parents believe that there will be fewer guests anyway. The cocktail party is to last for 4 hours.
- Complete the table to find the total costs for 320 guests.
 - How much will it cost per guest:
 - if all 320 people come?
 - if only 250 people come?
 - if only 100 come?



Item	Cost	Cost for 320 guests
Room hire	\$1000	\$1 000
4-hours of canapés	\$100 per person	
Drinks	\$50 per person	
1 wait staff	\$35 per hour	
DJ for music	\$100 per hour	

- 7** The cost of La Courgette is too much, so a compromise is sought. It is decided to have a party for 320 guests in a marquee (tent) with cheaper finger food and no wait staff. The students still want to provide the music and the hospitality students will act as wait staff.
- The parents are told they should allow an area of 0.5 m^2 per person in the marquee. What area of marquee should they hire?
 - If a 15 m by 10 m marquee were hired, how many fewer guests would this allow?
 - Two of the parents own a hotel, so they can provide 'gold star' finger food and a cheap drinks package. The parents will also provide all the plates, cutlery and glasses for free. Complete the table, right, to find the total cost for 320 guests.
 - How much will it cost per guest:
 - if all 320 people come?
 - if only 250 people come?
 - if only 100 come?

Item	Cost	Cost for 320 guests
Marquee hire	\$1350	\$1350
'Gold star' finger food (18 items each)	\$27 per person	
Drinks package	\$23 per person	
Wait staff	\$0 per hour	
Music students	\$0 per hour	

The world over, yachting is considered to be a thrilling sport. A series of yacht races is called a regatta. Regattas are big events that require a great deal of organisation. In this unit, plans are to be made for a 120-km yacht race around the Greek Islands.

Safety boat

- 1 There must be a safety plan for any yacht race event. First aid, fresh water, fuel, life-saving equipment and the weather need to be considered. There is usually a safety boat, which is the only motorboat allowed near the yachts!
 - a In the race, 12 yachts are expected to each carry 4 people. How many people will be sailing around the Greek Islands?
 - b If each person consumes 750 mL of fresh water every 4 hours, how many litres of fresh water are consumed by the sailors during an 8-hour journey?
 - c The safety boat usually has a 2-stroke motor. For this motor, the fuel mix is made from 1 litre of oil to every 50 litres of petrol.
 - i If 50 litres of petrol are used in the fuel mix, how many litres of boat fuel are produced?
 - ii How much oil in mL is required to make 21 litres of fuel mix?
 - d If 1.67 litres of fuel enables the safety boat to travel 1 km, how much fuel is required for the 120-km race?

Sailing through the numbers!

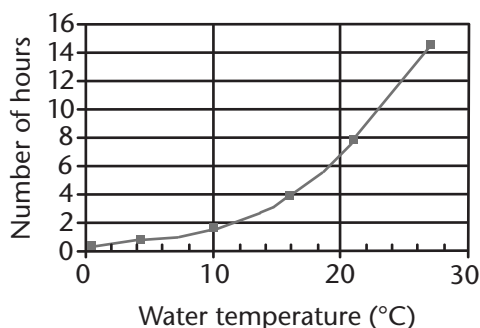
The managers of the race require the organisers to plot the positions of the yachts on a chart as they sail through the course. The race starts at Delos, heads to Mykonos, then Naxos and then back to Delos. The charts are in nautical miles, so plotting the sailing course often requires the conversion of kilometres to nautical miles.

... FAST MOVING FACTS

1 nautical mile = 1.852 kilometres
 1 knot = 1 nautical mile per hour
 1 knot = 1.852 km per hour
 1 km per hour = 0.540 knots

- 2
 - a Use Fast Moving Facts to convert 120 km into nautical miles. Give your answer correct to 3 decimal places.
 - b Speeds are expressed as nautical miles per hour or knots. The yachts average a speed of 10 knots. What is 10 knots in km per hour?
 - c How long will the race take if the yachts average 10 knots?
 - i Give the answer in hours to 4 decimal places.
 - ii Give your answer in hours and minutes.
- 3 In the rehearsal for the big event, the safety boat is 5 nautical miles away when it receives a 'mayday' emergency call from *Dee's Devil* reporting that one of the crew has fallen overboard. The temperature of the water is estimated to be 6°C, and the safety boat can travel at a speed of 20 knots. Use the graph to find if the crew member will be unconscious when the safety boat reaches him.

Time to unconsciousness



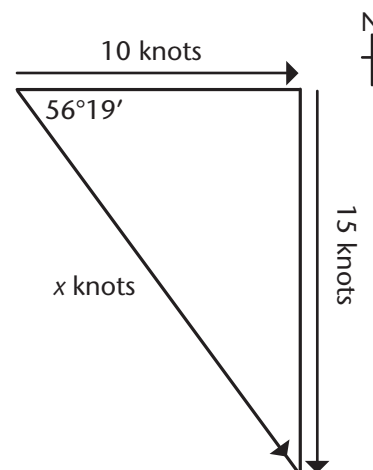
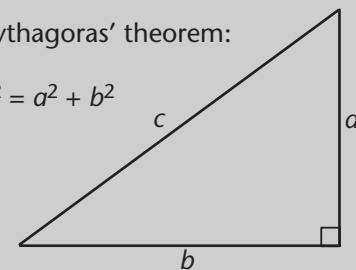
Okay, let's talk knots!

- 4 a It is clear that the yachts will have a headwind, which will slow them down. If the average speed of the fleet was only 5 knots, how long would the 120-km race take? Give your answer to the nearest whole number hour.
- b For part of the race, the safety boat will be motoring due south at 15 knots (see diagram, right), but the tide at that time will be running at 10 knots due east. This means that the boat will actually be travelling in the direction shown as x knots in the diagram. Use Pythagoras' theorem to calculate the actual speed, correct to 2 decimal places.
- c If the angle of the safety boat's course is as shown, which of the following would be the actual compass direction for its course?
- S56°19'E
 - S33°41'E
 - S33°41'W
 - N56°19'E

... FAST TRIANGLE FACTS

Pythagoras' theorem:

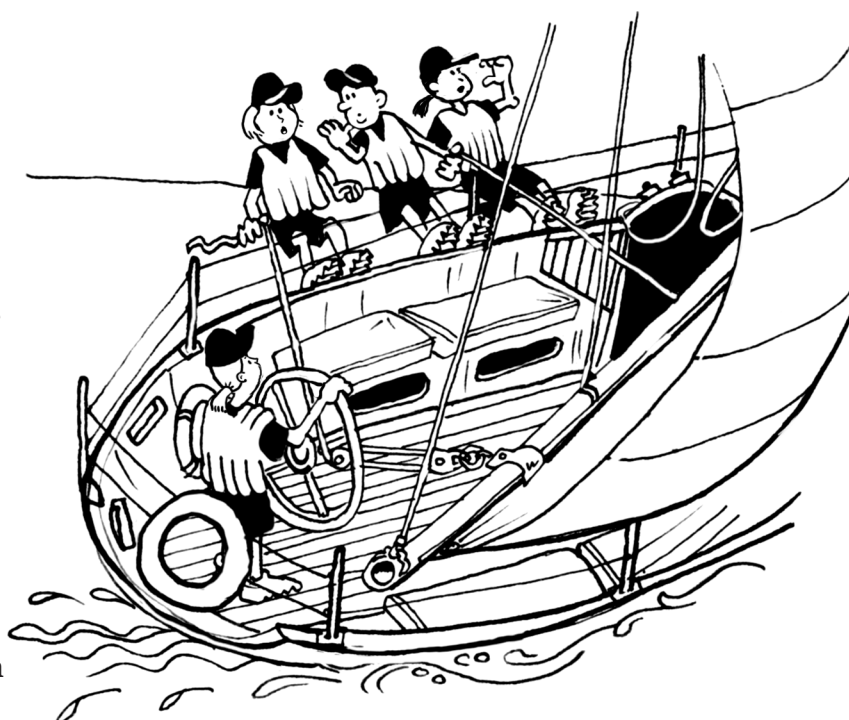
$$c^2 = a^2 + b^2$$



Buoy, that was some race!

The event was managed so well that no boats were lost and all the boats crossed the finishing line in time for the celebrations.

- 5 In the home stretch, between Naxos to Delos, the Greek vessel *Hellenic* was passed by the New Zealand vessel *Kiwi*, which went on to win by 0.5 km.
- By how many nautical miles did *Kiwi* win? Give your answer correct to 2 decimal places.
 - The winner crossed the line at 4.24 am and the second placed yacht, *Hellenic*, finished at 5.12 am. What was the time difference between the first two yachts?
 - The wooden spoon, the prize for the last boat over the line, was won by *Beejay*, which finished 3.3 km behind the winner. What was the average distance in metres between each of the 12 boats?



See-More High School in Melbourne, Australia, is arranging a Science study tour to the Great Barrier Reef, off the north-eastern coast of Australia. Twenty Year 12 students along with two teachers will fly between Melbourne and Cairns airports. The objective of the study tour is to investigate and report on a series of coral formations and other marine life.

Ready, set, jet!

The flights between Melbourne and Cairns will be aboard the AirAustralia airline.

- 1
 - a The plane is part of the 737-500 series fleet. The total passenger seating capacity for the aircraft is 200, 15% of which are seats allocated to Premier Class. How many seats on board the vessel are in Economy Class?
 - b The usual Melbourne–Cairns return airfare in Economy Class is \$150, including taxes. The Economy seats on the flight are fully booked. Use your answer to Question 1a to calculate AirAustralia’s revenue from this flight’s Economy Class fares.
 - c Premier Class seats on the Melbourne–Cairns return flight cost an additional 30%. Calculate AirAustralia’s revenue from this class.
 - d Calculate AirAustralia’s sales for the round trip to Cairns for all seats.
 - e The estimated travelling time from Melbourne to Cairns is 3 hours at a typical cruising speed of 835 km per hour. Calculate the distance travelled for the round trip in kilometres.

... FAST MONEY FACTS

Revenue is the total amount of money received for goods or services, before expenses are taken out.

Return is the income or profit from a sale. This is often expressed as a percentage.

What do you mean I am overweight?

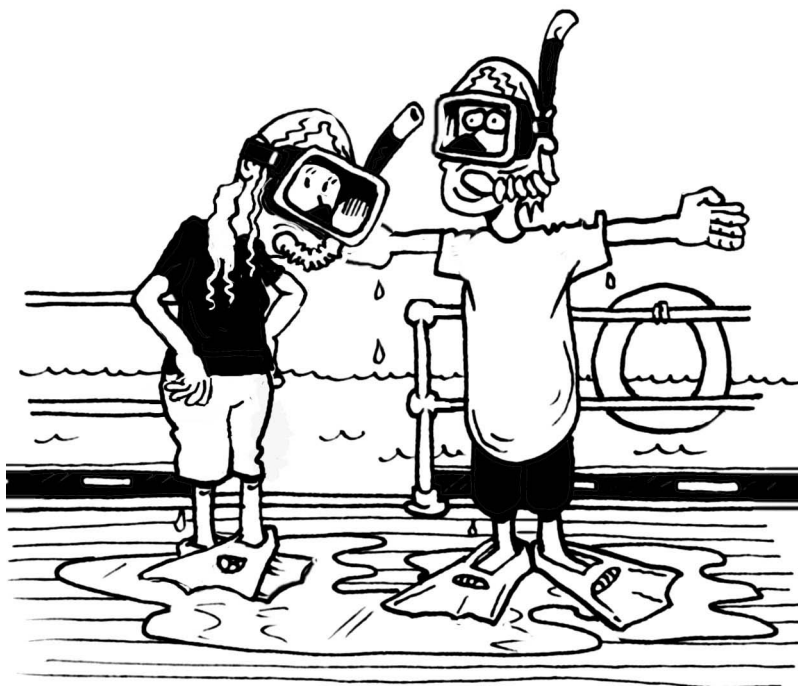
- 2
 - a Economy Class passengers on the Cairns flight are each allowed 20 kg of cargo luggage and 7 kg of carry-on luggage. If there are 170 Economy Class passengers, what is the maximum mass of cargo luggage for all Economy Class?
 - b What is the total mass of carry-on luggage permitted for all Economy Class seats?
 - c The Premier Class ticket price is 30% more than an Economy Class ticket of \$150. How much is a Premier Class ticket?
 - d Every kilogram that an Economy ticket holder’s luggage is overweight incurs an excess luggage fee of 5% of the cost of a Premier Class ticket. Two students, Jessica and Cameron, have packed too much! Jessica has an additional 4 kg and Cameron has an additional 5.5 kg. How much will they each have to pay for excess luggage if they are both travelling Economy Class?

The sleeper!

AirAustralia has introduced a new plane, the 777-300 series, to complement its existing fleet. This plane seats 350 passengers. Economy Class has 260 seats, Premier Class has 50 seats and First Class, renamed the ‘Sleeper’, has 40 seats. This new series costs 15% more for the Melbourne–Cairns return flight in Economy Class than the \$150 charged on the old 737-500 series.

- 3
 - a Calculate how much AirAustralia could receive in return fares if all the Melbourne–Cairns Economy Class seats on the new 777-300 series are sold.

- b** What is the Premier Class seat price if it is 35% more than the new Economy Class price?
- c** Hence, calculate how much AirAustralia would receive if all the Melbourne–Cairns Premier Class seats on the new 777-300 series were sold. Give your answer to the nearest dollar.
- d** The Sleeper seats cost 120% more than the Economy Class seat price. How much does this seat cost?
- e** Hence, calculate how much AirAustralia would receive if all the Melbourne–Cairns return flight seats on the new 777-300 series were sold.



Putting the tour together!

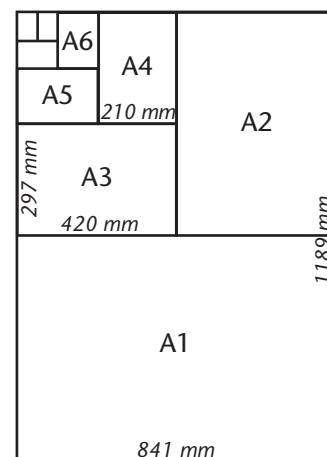
The costs of the various parts of the 6-day 5-night tour package are shown in the table, below right.

- 4** For the 20 students, there is a 10% student discount on the published airfare and then a further 5% group booking discount. The two teachers receive just the group booking discount. Use the table to answer the following questions.
 - a** Calculate the return airfare per student.
 - b** Calculate the return airfare per teacher.
 - c** How many hotel rooms are required for the 20 students?
 - d** Calculate the accommodation cost per night per student.
 - e** What is the total accommodation cost for the whole 6-day 5-night tour per student?
 - f** Calculate the total accommodation costs for the 2 teachers for the whole tour.
 - g** What will the total SCUBA dive cost be per student?
 - h** Calculate the meals cost for 2 meals on the first day, 4 days of 3 meals and 2 meals on the sixth day.
 - i** Calculate the out-of-pocket expenses for 5½ days.
 - j** Using your previous answers and the table, find the total cost per student for the tour.
 - k** Using your previous answers and the table, find the total cost per teacher.
 - l** Hence find the total cost for 20 students and 2 teachers to participate in the Science study tour.

Item	Cost
Return airfare Melbourne–Cairns, including taxes	\$150 per person
Both airport–hotel transfers	\$12.50 per person
Travel insurance	\$25 per person
5 nights accommodation (4-share, students)	\$120 per room per night
5 nights accommodation (2-share, teachers)	\$100 per room per night
1 day of SCUBA diving safety instructions	Free
3 days of guided SCUBA diving, including equipment (students and teachers)	\$50 per day
1 day to discover Cairns	Free
Meals (students and teachers)	Averaged at \$12 per meal
Out-of-pocket expenses (students and teachers)	Budget \$30 per day

Aaron has just been appointed as Event Manager for the Tivoli cinema chain. To be successful as an event manager, he must promote the 8 cinemas as venues for business functions, children's parties and social groups. He decides to produce a folder of information that he can display at the cinemas and give to prospective customers. He must work out the contents of the folder and how much the promotion is going to cost before his boss will let him go ahead.

- 1 There is an international standard for sizes of paper, and these paper sizes have the same ratio of height to width: $\sqrt{2}$. The basic size is a sheet of paper measuring 1 m^2 , which is called A0. The next paper sizes in the series are called A1, A2, A3, etc., and are made by halving the preceding paper size parallel to its shorter side. In the diagram, right, A0 is the whole sheet, measuring 1189 mm by 841 mm.



- Find the ratio of height to width for A0 paper, correct to 3 decimal places.
- Find the dimensions of A2 paper and find the ratio of height to width. Give your answer correct to 3 decimal places.
- Based on your answer to Question 1b, find the dimensions of A4 paper.
- An A0 sheet of paper measuring 1 m^2 weighs 80 grams. Express the area of an A4 sheet as a fraction of the area of an A0 sheet and thus find the weight of an A4 sheet of the same paper.
- An A4-sized picture is reduced to A5 size. What is the percentage reduction in area?



- 2 To advertise movies available for business functions, Aaron will include one A4 leaflet inside each of his 500 folders. He will use 80 gsm paper. Use Fast Paper and Card Facts and the area of an A4 sheet to find the mass of the 500 sheets.

- 3 The folder is made of 300 gsm cardboard folded in half so that the folder is A4 size.
- What are the dimensions of the cardboard used to make a folder?
 - What area of cardboard is needed for a folder in:
 - mm^2
 - m^2
 - What will be the weight of a folder? Give your answer correct to 2 decimal places.

... FAST PAPER AND CARD FACTS

Paper and card come in different masses. These masses are measured in grams per square metre (gsm).

- Thin tracing paper is about 50 gsm.
- Normal writing or photocopier paper is usually 80 gsm. Paper that is 160 gsm is twice as heavy, and it is twice as thick if it is the same density.
- Anything over 200 gsm is called card or cardboard.
- 500 gsm is thick card.

- 4** Aaron asked for a quotation from Walnut Design for the design and printing of a brochure to go in the folder. Use the quotation, right, to answer the following questions.
- What is the length and width of the pamphlet?
 - Is the cover made of paper or cardboard? Why?
 - Are the inside pages thick paper or normal paper?
 - How much will it cost to design, produce, prepare and print 500 copies of the pamphlet, without GST?
 - What is the cost when GST is included?
 - What will be the total cost if Aaron needs Walnut to make 2 hours of corrections? Don't forget to add GST.

Walnut Design, Quote 413

Specifications

Size: A5

Cover: 4 pages in full-colour on silk matt 300 gsm

Inside: 12 pages printed in black on silk matt 150 gsm, saddle-stitched

Delivery to be included

Design and produce:

12 pages plus cover – \$1600

Preparation and printing:

500 copies – \$2070

Price does not include 10% GST

Corrections will be charged at \$90 per hour

- 5** Aaron's business card will go inside each folder so that businesses that wish to book an event know how to contact him. He needs to decide what paper quality he would like for 500 copies of 86 mm × 54 mm business cards. He has decided on two colours on 360 gsm glossy card, and he collects the following information on the Internet.

Company	Quantity	Description	Total price, including GST and delivery
Webcard	500	Basic un laminated 310 gsm, glossy, one side only	\$49
Webcard	1 000	Basic un laminated 310 gsm, glossy, one side only	\$89
Artboard	1 000	Gloss front, 360 gsm, full-colour printing, free artwork	\$99
Spot Cards	1 000	Full-colour printing, 300 gsm, single-sided	\$115
Spot Cards	5 000	One-colour printing, 300 gsm, single-sided	\$60
Premium	500	Premium, one-colour printing, 350 gsm, single-sided Plus artwork	\$88 \$49
Premium Plus	500	Gloss one side, four-colour printing, 350 gsm, single-sided Plus artwork	\$120 \$49

- Which has the cheapest cost per business card?
- Which has the most expensive cost per business card?
- Can Aaron buy the card that he wants from these companies? If so, which one?
- Spot Cards offer him 25% discount on 1000 cards. Does this make their full-colour cards the cheapest buy per card?

The Alpha-Clubs Championship was held at the Aurora Stadium last night. It was an exciting game and the sponsors and both clubs were very happy with the event.

1 The Aurora Stadium was the first choice for the organisers of the championship for the reasons below. Use the calculations to find out the facts about the stadium.

- a** It can hold $(9 \times 6 - 1) \times 10^3$ fans.
- b** There are $5^2 \times 10^2$ undercover car parks available.
- c** The area of the arena is $2^3 \times 5^2 \times 10^2$ square metres.
- d** There are $\frac{10^2 \times 10^3 \times 10^4}{10 \times 10^2 \times 10^3}$ television monitors for business guests.
- e** $(5^2 - 3 \times 2) \times 10^6$ people have attended the Aurora Stadium since it opened in the year 2000.

... FAST SOCCER FIELD FACTS

The official dimensions of a soccer field are:

- length: between 90 m and 120 m
- width: between 45 m and 90 m

For international matches, the official dimensions are:

- length: between 100 m and 110 m
- width: between 64 m and 75 m

2 Use the Fast Soccer Field Facts and the area of the Aurora Stadium arena from Question 1c to find out if the area of the arena is likely to be large enough for:

- a** a local soccer match
- b** an international soccer match.

3 **a** The crowd at the Alpha-Clubs Championship game was 53 273, in a stadium seating 52 500; the rest needed to stand. How many fans needed to stand?

- b** Regulations state that for every 5000 people or part thereof, the organisers need to provide 6 toilets, 25 urinals and 17 washbasins for men and 34 toilets and 17 washbasins for women. List the toilet facilities required for the game.

4 An overview of the key features of the game will be reported on the TV news. This will be good advertising for the event.

- a** The Alpha-Clubs Championship game began at 7.30 pm. Both teams finished the match 90 minutes later, each with only 10 men on the field. At what time did the game finish?
- b** In the 20th minute of the match Billy, playing for the Reds, scored the only goal of the match. At what time did Billy score the goal?
- c** In the 10th minute of the match Angelo, playing for the Blues, was sent off after hitting Georgio, who was playing for the Reds. Then Georgio was sent off at the 65th minute. For how long did the Reds have more players on the field than the Blues?



- 5 The Aurora Stadium has a number of function rooms used for different activities, as listed in the table, below. Use the table to answer the following questions.
- Which area will allow the greatest number of guests in a theatre function?
 - Which room would need to be used for a banquet for 1160 guests?
 - Which area would be best for a small banquet of fewer than 50 people?
 - The Captain Cook and Parkview Rooms can be used together and laid out for cocktails for corporate events. How many guests can be entertained in this way?

... FAST 'GROSS' FACTS

Gross means before any expenses or taxes have been taken out.

Gross income will include gate receipts, advertising, sponsorship, etc., before expenses have been taken out.

Gross receipts from sales is the amount received before expenses have been taken out.

Area	Theatre function	Cabaret function	Banquet	Cocktail function
Peak Room	200	250	300	30
Arena Viewing Room	60	30	50	70
Captain Cook Room	200	120	250	300
Parkview Room	250	100	300	500
Club Brasserie	500	350	1 200	n/a
Club Terrace Bar	150	120	400	n/a
Total	1 360	970	2 500	900

- 6 The cost of hiring the Aurora Stadium for a football match is \$1 500 000 or 20% of the gross income, whichever is higher. How much would be charged for hiring Aurora for the Alpha-Clubs Championship if the organisers received on the average \$100 gross from each of the 53 000 attendees?

- 7 There are a number of extra costs when hiring Aurora Stadium (right). The organisers of the Alpha-Clubs Championship feel that they will need exclusive use of the TV monitor, use of the scoreboard and a sound technician. They will handle the sale of souvenirs and the distribution of programmes themselves, but the stadium insists on charging 25% commission on the \$5 300 000 gross receipts for the merchandising and programmes. How much will the organisers of the championship pay for 'extras'?

Optional extras	Charge (\$ per day)
Ground TV monitor (exclusive use)	\$84 000
Ground TV monitor (shared use)	\$29 000
Scoreboard	\$5 000
Sound technician	\$5 000
Souvenir shop	\$5 000
Programme distribution	\$11 000
Extra chair (max. 1000)	\$4
Sale of merchandise and programmes	25% of gross receipts

- 8 The coaches wanted their teams to get used to playing in the Aurora Stadium. Each of the two sides had two 2-hour sessions during the day without spectators and one 2-hour session at night under lights with their fan clubs. See the charges, right. How much did Aurora Stadium receive from the clubs for their training sessions?

Sports training hire charges	Amount
Two-hour session, without spectators	\$8 540
Two-hour session, with no more than 300 spectators	\$10 450
Extra: use of floodlights (cost per hour)	\$1 860



The Sail Past, or Review of the Fleet, usually marks the start of the boating season. It is a big event because it is part of 600 years of naval tradition and history. Sometimes there is also a Blessing of the Fleet, like those used to call for good fishing and a safe return for fishermen.

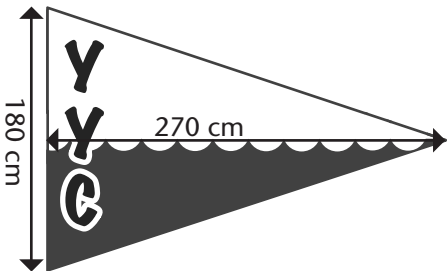
Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

Event day

The annual Yanchep Sail Past and Blessing of the Fleet is being organised for Saturday 3 October to mark the opening of the racing calendar. It is to be hosted by the person in charge of the club, called the Commodore.

Time	Activity
12:00 hours	Skippers' meeting
12:20 hours	Raise the club burgee up the flagpole
13:00 hours	Boats leave the docks to assemble in the marshalling area, downwind of the Commodore's boat
13:45 hours	Sail Past and Blessing of the Fleet signalled with one loud signal. The signal is preceded by a warning signal and radio call at 13:30, 13:35, 13:40 and 13:44
17:00 hours	Commodore's reception
18:00 hours	Formal dinner

- 1 The timings of the event are shown in the table, right. Timing the Sail Past is very important to the success of the event. My 3-m yacht, *Beejay*, can sail at 8 km per hour.
- a How long do I have to sail to the marshalling area?
 - b If the marshalling area is 5 km from my berth:
 - i how long, in minutes, will it take *Beejay* to sail there?
 - ii will *Beejay* make it in time for the start?
 - c How many warning signals and how much warning time is there before the start of the Sail Past?
 - d If the formal dinner ends at 10.30 pm, how long is it from the skippers' meeting until the end of the night's celebrations?



Raise the burgee!

A burgee is a club's flag, usually triangular, which can be flown on the flagpole at the club or on a yacht when it is sailing or anchored.

- 2 The club needs new burgees for the event.
- a The first one is for the clubhouse. The club chooses a large one, 180 cm by 270 cm. What is the area of the flag in cm^2 ?
 - b The price of the flag is \$175, plus an artwork cost of \$45. Find the cost of the flag.
 - c What is the cost of the flag per square metre?



- 3 The club will also purchase some burgees for the yachts to hoist for the Sail Past. Because they will buy in bulk, there will be no fee for the artwork. See the order form, right.

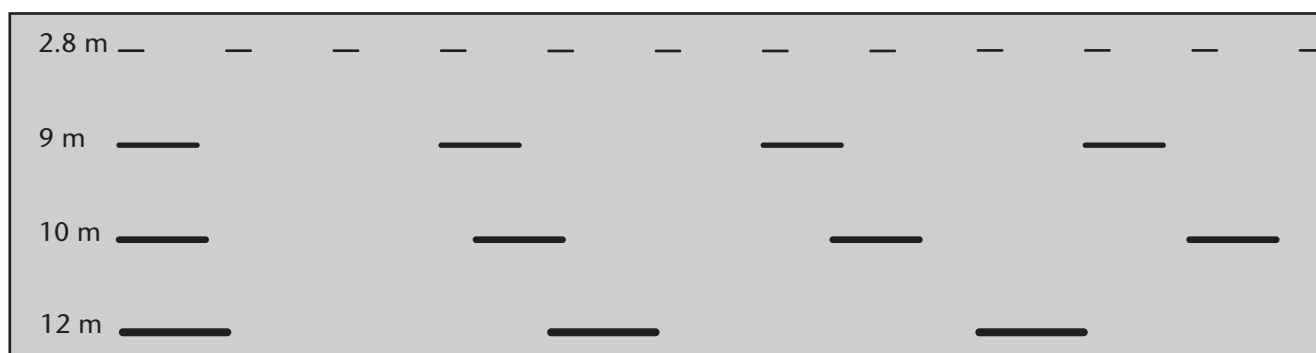
- a How much will it cost the club altogether?
- b If they sell all the flags to club members at a profit of 12.5%, how much profit will the club make?

Order form for Yanchep Yacht Club

No.	Type	Size cm ²	Price	Total
15	Dinghy, 2.8 m	250 × 450	\$18	
24	6–9 m	300 × 450	\$24	
19	7.5–10 m	300 × 450	\$30	
7	9–12 m	360 × 530	\$36	
			Total	

Aye-aye, Captain!

- 4 Each yacht will salute the Commodore of the club by releasing its jib, or the sail at the front, and letting it flap. The salute commences one boat length before and continues for one boat length beyond the Commodore's yacht. For how long will a 12-m yacht have its jib flapping?
- 5 The organisers of the event must ensure that all boats sail in *a single line* and keep 3 boat lengths between them and other yachts during the Sail Past. The picture below shows how each of the groups of yachts will sail past. The 2.8 m dinghies will be $3 \times 2.8 = 8.4$ m apart, followed by the 9 m yachts at $3 \times 9 = 27$ m apart etc.



- a How far must the 10 m and 12 m yachts keep between them?
- b First there are 15 of the 2.8-m dinghies, then 24 of the 9-m yachts, then 19 of the 10-m yachts and then 7 of the 12 m yachts in the Sail Past. Approximately how long will the whole Sail Past be? Choose one of the following answers:

- A 750 m B 1.7 km C 2.3 km D 3.4 km

- 6 a The order of the committee yachts is as follows: Past Commodore, followed by Vice Commodore, Rear Commodore, Fleet Captain, Treasurer and Secretary of the yacht club. If the order was random instead, in how many different orders could the committee yachts sail past?

Idea: $6 \times 5 \times 4 \times 3 \times 2 \times 1$

- b The order of the club yachts must be as follows: 7×12 -m yachts, 19×10 -m yachts, 24×9 -m yachts and 15×2.8 -m dinghies. If the order of the yachts within the groups was random, in how many different orders could the club yachts sail past?

Idea: $\frac{65!}{15! \times 24! \times 19! \times 7!}$ where ! is called 'factorial' and means, for example,

$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$ and $4! = 4 \times 3 \times 2 \times 1$ and $3! = 3 \times 2 \times 1$



The WonderDome is Australia's premier multi-purpose Entertainment Arena. In 2015, Australia is hosting the Golden Racquet Tournament (GRT). This is a 10-day event, for which players, family, managers, media and guests will fly in from all over the world. The event is being managed by the GRT EventsInc Group.

Ready, set, deuce!

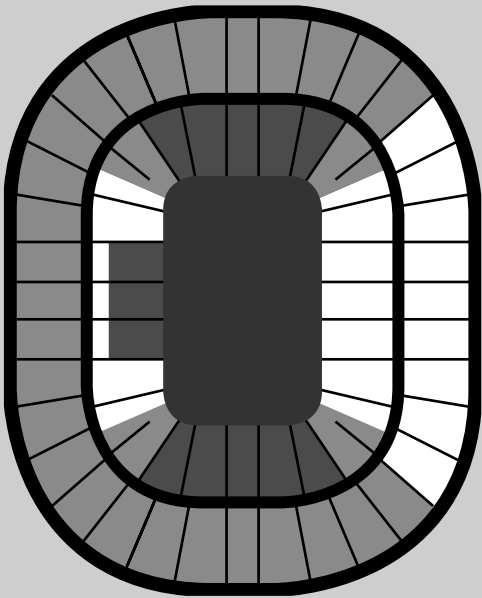
- 1 The WonderDome has a seating capacity of up to 12 000 people. Use the seating map and the ticket price table in Fast WonderDome Facts, to answer the following questions. Okay, the ball is in your court ...
- a Calculate the ticket sales for each section. Complete the table, below.

Section	Ticket sales (\$)
A. Light grey (outer)	
B. Light grey (inner)	
C. Dark grey	
D. White (outer)	
E. White (inner)	

- b What are the total ticket sales for all sections?
- c What are each section's ticket sales as a percentage of the total sales? Complete the table, below.

Section	% of sales
A. Light grey (outer)	
B. Light grey (inner)	
C. Dark grey	
D. White (outer)	
E. White (inner)	

... FAST WONDERDOME FACTS



Section	No. of sections	No. of seats sold	Unit price per day
A. Light grey (outer)	27	5 400	\$220
B. Light grey (inner)	4	600	\$280
C. Dark grey	13	180	\$360
D. White (outer)	9	150	\$460
E. White (inner)	7	200	\$580

- d The tournament runs for 10 days. Calculate the WonderDome's total ticket sales for this period, if the amounts in the table are average daily sales.
- e If 20% discount were to be offered for a 10-day pass, how much would a pass cost for:
- i Section C?
 - ii Section E?
- f The WonderDome calculated that its cost to stage the event would absorb 64.5% of the total ticket sales for the 10 days. Find the value of the profit for the event.

Follow the bouncing ball!

- 2** GRT EventsInc Group negotiates with the WonderDome's court hire manager regarding tennis court hire costs. The WonderDome has 3 show courts that are hired together by the hour. There is a day and an evening rate. The 10-day tournament is divided into two main stages that have different hours: a 7-day stage for the tennis heats and a 3-day stage for the finals.

- Use the Fast Court Facts to calculate the total hire cost for:
 - the first 7 days of play?
 - the last 3 days of play?
- Hence, find the total court hire cost for the tournament.
- How many hours of play will there be for the 10-day period?
- What is the average cost of court hire per hour for the tournament?

... FAST COURT FACTS

Court hire hourly rates

7 am–5 pm: \$38 per hour

5 pm–11 pm: \$45 per hour

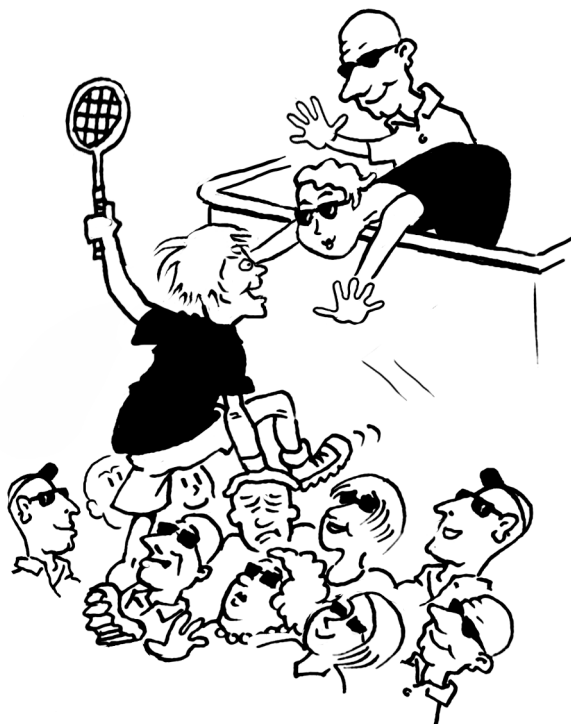
Session times

Heats: 7 days: 7 am–4 pm, 6 pm–10 pm

Finals: 3 days: 7 am–6pm, 7 pm–11 pm

Super-size me!

- 3** The WonderDome features 12 Super Boxes that are usually hired by corporations to entertain special local and overseas guests. The hire cost per Super Box for a 10-day period is \$40 000. Each Super Box comes as a package. The package's features are listed in the box, below right.
- Using the 10-day deal, calculate the cost to set up a Super Box for hire.
 - Calculate the total revenue for the 12 Super Boxes.
 - Calculate the profit of the 12 Super Boxes for the 10-day tournament.



Super Box - 10-day deal

25-seat capacity per box

Package includes:

- 5 premium car parks, valued at \$200 each
- Personalised passes, valued at \$10 per head
- Catered lunch, valued at \$27.50 per head per day
- Catered dinner, valued at \$37.50 per head per day
- Service staff for 40 hours, valued at \$30.00 per hour
- Security, valued at \$3000

Promoters are organising an annual event in Australia called Fashion Week. This important event is to promote existing and new fashion designers to buyers from all over the world. During Fashion Week, parades are conducted and display stands are fitted out to best show and promote fashion collections. This year's Fashion Week is themed 'Botany', as a way of emphasising the importance of the environment. Eco-friendly products are used throughout the exhibition centre.

Showing off!

The promoters of Fashion Week are one of Italy's main fashion houses, called Fila, meaning 'thread' in Italian. The WonderDome is hosting this prestigious event. Six runways allow fashion houses to display their designs.

- 1
 - a Promoters are expecting 12% more people than at previous events. The previous Fashion Week drew 250 000 people. Calculate the number of people expected this year.
 - b At the previous Fashion Week, \$50 000 000 worth of fashion purchases were made, but increased awareness is expected to boost sales by 6% this year. What could be the new fashion purchase amount?
 - c Four runways, each with a seating capacity of 1000 guests per day, and two runways, each with a capacity to seat 600 per day, are available. Calculate the total number of guests that could sit across all runways per day.
 - d Calculate how many seated guests are possible across all runways over 7 days.
 - e Using the answer to Question 1b calculate the anticipated average whole dollar spend of each of the 4500 fashion buyers.

Take a stand on fashion!

A total of 1500 display stands at the event are hired to fashion brands, caterers, magazine sellers, forecasters of fashion trends and first aid providers.

Display stand options	No. of stands available	Hire fee, per stand	Total hire fee	Total floor space
Option A: 1.3 m x 3 m	1 000	\$2 000	\$	m ²
Option B: 2.6 m x 3 m	500	\$3 500	\$	m ²

- 2
 - a The promoters calculate some of their income from the stands they hire out. The stand sizes are indicated in the table. What is the average price for a display stand?
 - b Using the table, calculate the potential total hire fee for both display stand options.
 - c Using the table provided, calculate the total floor space each option occupies in m².
 - d What is the total potential revenue from the stands?
 - e Calculate the total display area for Fashion Week.
 - f In Question 1b, the potential invoice amount was calculated at \$53 000 000. Use this total display area in m² to calculate the average return on each m².

Runway chic!

- 3
 - a Each show comprises 6 brands and runs for 30 minutes. The shows are staged with 1½ hour intervals and are timetabled between 10 am, for the first show, until 6 pm, for the last show. How many shows run between 10 am and 7 pm?

- b** How many shows run during Fashion Week?
- c** The cost to each designer is \$4000. If one runway show parades 6 brands, how much income comes from:
 - i** one runway show?
 - ii** a day of parades?
 - iii** all the runway shows?

The colour of botany!

Fila is supplying the main signage as well as various plants to ensure the 'botany' theme is reflected throughout the WonderDome. Fila also handles the invitations and all the promotional activities for the event.

- 4 a** A security team, caretakers, gardeners, cleaners and a medical unit are on site for the week. These services are charged as an extra fee, along with the stand hire fees. This extra fee is 5% of the stand's hire charge. Using the table in Question 2, calculate the total income from this extra fee.
- b** The total advertising and promotional cost for the event is approximately \$555 000. What is the average price of advertising per display stand?
- c** What is the amount spent on advertising and promotion per m²?

The catwalk!

- 5** Fashion designers make money from Fashion Week through publicity, runway shows and their exposure to local and overseas fashion buyers. Use Fast Catwalk Facts to answer the following questions.
 - a** There are 6 male and 10 female models booked for the event. What is the total model hire cost for the week?
 - b** The promoters are flying in Couturia, a European supermodel. Her fee for parading in 7 key runway shows is 4 times greater than the average specified in the list. What is the



... FAST CATWALK FACTS

All prices include GST, agency fees and all assistants

- Models: ~\$4000 each, for 7 days
- Fashion stylist: ~\$7000 for 7 days
- Make-up artist (4 needed): \$90.00 per hour each
- Hair stylists (4 needed): \$120 per hour each
- Shoes for models: \$250 per pair

approximate whole dollar cost for Couturia to parade in each show?

- c** Make-up artists are required for 1½ hours per show. What is the total cost for them across 35 shows in the week?
- d** All hair stylists are required for 1½ hours per show, for 20 runway shows. What is their total cost?
- e** Each model has 4 shoe changes in a runway show. How many pairs of shoes in each show?
- f** What are the shoes going to cost?
- g** Calculate the final costs to the promoters for the 16 models, plus Couturia, the fashion stylist, make-up artists, hair stylists and the shoes.



Roberto and Brigitte have planned their wedding well in advance because the venue needs to be booked at least one year ahead. They have decided to give their wedding an Italian–French theme in keeping with their families’ traditions. They must arrange the clothes, flowers, cars and the rings. Keeping a running total will help you answer the final question.

The reception

The reception will be held on the 20th floor of the Style building, located at Sydney’s waterfront. The circular dining room, Dome, will comfortably seat the 302 people at the reception at 24 tables.

Menu at Dome

Three-course dinner for \$175.00 per adult

- Italian antipasto
- Duck confit
- Crème caramel
- Champagne, wine and soft drink

- 1 a The bride and the groom have decided that each is to pay for themselves and their own guests. Roberto is paying for 195 people and Brigitte is paying for the rest. What are the costs for:

i Roberto?

ii Brigitte?

b Twelve of the guests are under 12 years of age and Dome will charge them only \$125 each. What then will be the total cost of the 302 dinners at Dome?

Do I look like a meringue in this dress?

- 2 a Brigitte calculates the cost of outfitting the female bridal party. She writes all the costs into a table.

i Brigitte has seen the dress she wants. The dress has a base cost of \$3200, plus 5% of the base cost for each visit to the dressmaker for adjustments. Work out how much the dress will cost.

ii The veil is an additional cost. Use the information in the table to work out how much it will be.

iii Brigitte’s shoes cost \$370. Calculate the cost to dress the bride completely on her wedding day.

b i According to information in the the table, how much will each bridesmaid’s dress cost? Hence, how much for all the bridesmaids’ dresses?

ii The bridesmaids’ shoes are \$250 each. What will it cost to completely dress the 3 bridesmaids?

c Brigitte’s mother and future mother-in law’s outfits cost 10% more than the bridesmaids’ dresses. The maid of honour will wear a vintage French wedding dress.

i How much do the two dresses cost altogether?

Female bridal party costs		
a: Bride’s outfit		Total:
Base cost		
4 visits to dressmaker @ 5% of base cost		
Veil @ 30% of base cost		
Shoes		
b: Bridesmaids’ outfits		Total:
3 dresses @ 22.5% of base cost		
3 pairs of shoes @ \$250		
c: Mother’s and mother-in-law’s outfits		Total:
2 dresses @ 10% more than bridesmaids’ dresses		
2 pairs of shoes @ \$275		
d: Bouquets		
3 bouquets @ 10% of \$185		
e: Gifts		
7 pieces of jewellery @ \$450		
f: Total cost for female bridal party		

ii Both mothers' shoes cost \$275 a pair. What will it cost to completely dress both mothers?

d Brigitte wants three bouquets, one to throw to the female guests and the others as gifts to her mother and future mother-in-law. Each bouquet costs \$185. A discount of 10% is given for 3. What is Brigitte's total florist bill?

e Roberto buys the bride, bridesmaids, maid of honour, his mother and his future mother-in-law jewellery, each to the value of \$450. Calculate the cost to buy gifts for all the females in the bridal party.

f Hence, find the total cost for the female bridal party's clothes, flowers and jewellery.



Okay, what's for hire?

Roberto is organising his part of the wedding celebration. He has to buy the rings and the wedding cake; the rest he thinks he can hire.

3 a Roberto's ring cost 40% of the price of Brigitte's ring, which was \$2500. Calculate the cost of the rings for the bride and groom.

b The wedding cake will cost \$2.15 per guest. If there are 302 guests attending the wedding, calculate the cost of the wedding cake.

c Roberto needs to arrange suits for 3 groomsmen, his best man, his dad, the future father-in-law and himself. He has chosen black dinner suits with white shirts and white ties. The lapel flowers work in with the bride's bouquet. Using the spaces provided in the table, right, calculate the total charges for all the items.

d How much will it cost Roberto for all the hiring?

e What are Roberto's total expenses, including the cost of the rings and the cake?

Product/service	No.	Hire or purchase fees	Total charge
Car hire, per car per day	3	\$235.00	
Suit hire, per outfit	7	\$110.00	
Photography package	1	\$3 500.00	
Photographer's assistant (extra 10% of package)	1		
Hairdressers, per day	2	\$250.00	
Stylist	2	\$275.00	
Invitations, including design, printing and postage	140	\$3.10	
The band	1	\$1 500.00	
DJ (30% of band hire fee)	1		
Master of Ceremonies	1	\$360.00	
Flowers for suit lapels	7	\$8.50	

4 Okay, how much is the entire wedding going to cost including dinner, clothes, rings, hiring and products and services?

Staging an art exhibition is often a major event for an artist, a gallery and a community. Art galleries are generally located in major cities and key regional centres. Important shows are often exhibited internationally as a way of exchanging ideas between countries.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

Ivana Paintalot

Gallery Praxis in Sydney, Australia, has negotiated with modern artist Ivana Paintalot to exhibit at their gallery for their June show.

- 1
 - a A catalogue showcasing Ivana's work has been produced, including an essay about her work by a specialist writer. The cost price to the gallery for each catalogue is \$9.50. The gallery sells each for \$14.95. What would the profit be if 500 were sold?
 - b There are 16 two-metre square oil paintings and 12 one-metre square charcoal drawings. The retail price for the two different forms are \$8250 each and \$3450 each respectively.
 - i Calculate the possible revenue to the gallery if all the paintings and works on paper are sold.
 - ii What percentage of the possible revenue is expected from the works on paper? Give your answer correct to 3 decimal places.
 - c
 - i All the works on paper were sold and the gallery charged 20% of the wall price to frame each of the works. How much does the gallery receive for the framing?
 - ii The cost price to the gallery to frame each work on paper is \$350. Calculate the gallery's profit on framing.
 - d
 - i If $\frac{3}{4}$ of Ivana's paintings and $\frac{2}{3}$ of the works on paper were sold, how much would the gallery receive for the sales?
 - ii Gallery Praxis pays the artist 60% of the retail price, excluding the framing charges. Calculate Ivana's payment on these sales.

... FAST GALLERY FACTS

- Government-run galleries usually own the works they exhibit.
- Privately-owned galleries charge to exhibit and sell the works on behalf of the artist or artists.
- The person who arranges the gallery space is called the **curator**.
- The retail price of a painting is referred to as the **wall price**.
- When inviting guests to an art exhibition, the gallery relies on its **mailing list**. The artist may also have their own list.
- **Commission** is a percentage of the selling price that the gallery receives for selling a piece of art.
- **Profit** = selling price – cost price

Rodney McClick

For the July show, Gallery Praxis is exhibiting Rodney McClick, a young modern photographer.

- 2 Gallery Praxis has approximately 800 people on its mailing list, of which 20% are overseas clients. The artist's list is 30% of the size of the gallery's list and everyone is local.
 - a How many invitations could be sent if the gallery combines its list with that of the artist?
 - b How many invitations are being sent to overseas patrons?
 - c Rodney McClick has produced 40 photographs of the Australian outback. The works are priced at \$1750 each. Rodney sells 70% of his images off the wall.
 - i How much was received from sales?
 - ii If the gallery pays him 60% of that value, how much money did Rodney earn?

- d** If another 30% of Rodney's images were sold as extra editions and the gallery makes 40% commission on these, calculate the gallery's commission on these additional sales.

The opening night of an exhibition is an important marketing event for the gallery and the artist. Generally 1 in 3 of the combined local mailing lists will attend the opening night. Knowing this helps plan the evening.

- e** Calculate the likely whole number of patrons attending the opening night.
- f** If approximately 5% of the guests attending the opening night purchase an image:
- i** what whole number of people will purchase an image?
 - ii** how much money would Rodney make after the gallery took its 40% commission?

Freight on board!

Gallery Praxis showcases artworks on their website for international promotion and purchase. The works are priced at \$1750 each.

- 3 a** Of the 160 overseas people on their mailing list, 15% purchased an artwork.
- i** How much money was received from the sales?
 - ii** How much did the gallery make if it receives 40% commission?
- b** A freight company will send 5 of the works to New York, 5 to London and 5 to Singapore. The freight rate to New York is \$110 per piece, 8% less than that for London and 15% less than the London figure for Singapore. How much will it cost the gallery to send the 15 works?
- c** What percentage of the gallery's commission does this international freight represent? Give your answer correct to 2 decimal places.

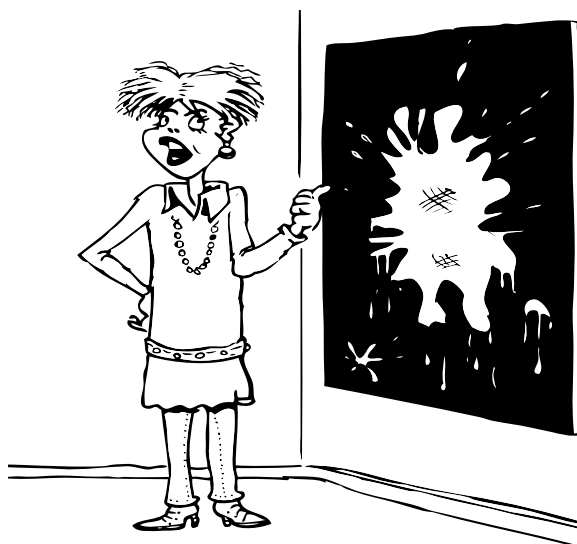
The Young Artist Award Dinner

Gallery Praxis and ArtAust have arranged a gala dinner, with entertainment and keynote speakers, to raise funds for the Young Artist Achievement Awards (YAAA). Government dignitaries, fine art institutes and Arts patrons have been invited, a total of 350 guests.

- 4 a** What was the total income raised by the gala dinner?

Idea: $14.9 \times \sqrt{144} \div 6 \times 7.5 \times 350$

- b** If the cost to produce the dinner and manage the event cost \$111.75 per guest, calculate how much money is left to fund the YAAA.
- c** Three awards are being offered: one for painting, one for sculpture and one for photography. The award for painting is \$10 000, the one for sculpture is 200% of that value and the award for photography is 25% of the award for sculpture. Calculate the value of each award.
- d** The balance of the money is donated to the YAAA fund. What percentage does that represent? Give your answer correct to 2 decimal places.



Street fairs and festivals are an exciting way for local communities to get together while promoting cultural diversity, music, film, food, and local arts and crafts. The budget for a street fair may be covered by the state government, the council and local shopkeepers, with the rest coming from sponsorship, donations and other activities.

Windsor Street Festival

- 1
 - a This year the 10-day festival of concerts, films and exhibitions has a budget of \$1.2 million. The event organisers have applied to the state government for 35% of the project budget. Calculate this figure.
 - b Local traders have agreed to contribute an equivalent of 80% of the government's contribution. How much money will come from the traders?
 - c The festival is over 4 street blocks in Windsor Street. It closes with a street fair, held from 8 am to 11 pm on the last Sunday. The 8 food stalls in each street block will be charged a rent of \$1680 for the day. What is the total rent money received for all 4 blocks?
 - d There will be a \$2.50 parking day pass for those who drive to the festival. Calculate the amount of money that could be raised if 2200 car places are provided every day.
 - e Calculate how much extra sponsorship money would need to be raised in order to meet the forecasted budget.

It's all for a worthy cause!

GLOSS hair products are offering a free haircare show and on-stage hair demonstrations in return for their logo appearing on all festival materials. GLOSS will be the festival's major sponsor.

- 2
 - a What is the cost to GLOSS if they provide 5000 free haircare showbags worth \$12 each?
 - b The organisers have decided that they need \$400 000 in sponsorship. GLOSS has agreed that their contribution will be 37.5% of this figure. Calculate GLOSS's total sponsorship contribution, excluding the showbags.
 - c Universal night club and the UDance club will each contribute 12% of the total sponsorship in exchange for their logos and advertisements being used in the festival brochure and on signs in the street. Calculate their total contribution.
 - d Radio4U2 will be exclusively covering the 10-day festival activities in various languages. They are expecting \$2.5 million of advertiser income, of which they have agreed to contribute 2% to the festival. How much money is that?
 - e Protect Technologies are providing 4 large digital screens (with their logo) playing music videos continuously for the day of the street fair, from 8 am until 11 pm. Their sponsorship is worth \$1500 per hour. How much money is that?
 - f How much more sponsorship money is left to raise?



Programming

A programme will be produced and slipped into the local paper so that people will know what the different events are and when they will be on. The graphic design students of a local university have arranged the layout and artwork for the programmes at no charge.

- 3 a** The cost to print one 12-page programme is \$6.25. A 4-page programme costs 60% of the cost of a 12-page programme. How much will it cost to print a 4-page programme?

- b** Usually there are special rates for printing big orders of event programmes. The festival organisers order 5250 of the 12-page programmes. Use Fast Printing Facts to calculate the printing cost.

- c** What would have been the printing cost if an extra 1000 of the 12-page programmes had been ordered.

- d** The organisers decide they will need 10 000 of the 4-page programmes to put in the newspapers. They are offered a printing cost of \$1.74 per programme.

i What percentage of the cost of a 12-page programme is this?

ii How much will these programmes cost?

- e** Calculate the total cost to the festival organisers if 6250 of the 12-page programmes are ordered along with 10 000 of the 4-page programmes?

- f** What would it have cost the festival organisers for all these programmes if they had not got the special prices for the big orders?

- g** Calculate how much money has been saved because of special prices for big orders.

... FAST PRINTING FACTS

Better prices on 12 pages for bigger orders!

Number printed	Price per program
2000–2499	\$4.50
2500–4499	\$3.80
4500–6499	\$3.23
6500+	\$2.90

Screens all right to me!

Windsor Street has a large cinema, called M-Screen, running for 5 days of the festival.

- 4 a** If M-Screen seats 1440 guests at each of the 5 films shown each day, calculate the number of people that M-Screen could seat for the period.

- b** The adult ticket price is \$8 and those under 16 years of age pay \$5 per ticket. If $\frac{3}{4}$ of the people found in Question 4a are paying the full price, how much money would be made from those 16 years of age and under?

- c i** Assuming an 80% capacity for each session is achieved (of which $\frac{3}{4}$ paid full price), calculate the amount of money returned from sales of tickets.

ii Hence, how much is paid to the festival organisers if they receive 7% of total sales?

- 5** A portable movie screen has been installed at the other end of the street to provide open-air movie screenings. The open-air movie shows will comprise 2 movies per night for 3 nights, half of which are foreign films. 650 people are expected at each film.

- a** Calculate the expected ticket sales for all the foreign films if all tickets cost \$6.50.

- b** \$1.50 will be paid to festival organisers from each ticket sold. Calculate how much money will be paid to the festival organisers if the expected open-air movie numbers are achieved.

A variety of merchandise is usually available as part of local, national or international events. This helps promoters and event managers to raise money to pay for special guests or for the event itself.

Badge of honour

A soccer club held a game in Brisbane for which satin badges were sold to help promote the soccer grand final.

- 1 The Club bought 10 000 satin badges for \$2.45 each and sold each for \$4. All the badges were sold. If the profit is the difference between the amount received from sales and the amount the badges cost, how much profit was made by the club altogether?

Some hair-raising sales!

To coincide with the opening of their renovated salons, the owners of Ecco Hairdressing wanted to thank their loyal customers with free tickets to the Three Divas Concert in the Park.

- 2 The event was attended by 120 lucky ticket holders. The tickets were priced at \$55 each.

- a How much did Ecco Hairdressing pay for all its tickets if they received a 20% discount on the selling price?
- b The venue holds 6120 people. How much did the concert organisers receive for tickets, taking into account Ecco's discounted tickets. The amount of money received altogether for tickets is called the 'gate'.
- c The Three Divas are paid 10% of the gate. How much did each of them receive?



- 3 GLAM Events designed a range of merchandise for the crowd at a Rave performance. The T-shirts, CDs and DVDs were sold at interval and at the end of the show. T-shirts were sold for \$17.50 each, CDs for \$20 each and DVDs for \$25 each. GLAM Events recorded 5200 people attending the event!
 - a If 40% of attendees bought T-shirts, calculate the amount of money made in sales from this product.
 - b If 45% of attendees bought CDs, calculate the amount of money made in sales from this product.
 - c If 60% of those who bought CDs also bought DVDs, calculate the amount of money made in sales from DVDs.
 - d Calculate the average amount per attendee that GLAM received from merchandise sales.

I saw, I volleyed and I bought the T-Shirt!

Over a 10-day tennis tournament, WonderDome Australia sets up booths selling event-related merchandise. The organisers arrange to sell T-Shirts, caps, tournament rackets and a pack of four tennis balls. All products bear the Golden Racquet Tournament logo.

- 4 The WonderDome has a seating capacity of 12 000 people and the company has forecast sales as a fraction of the total number of people. They use this information to order from the manufacturers. Use the fractions in the table to:

Fraction of people	Product	Retail price	Quantity	Total value
$\frac{1}{4}$	T-shirt	\$28.00		
$\frac{1}{3}$	Cap	\$15.00		
$\frac{1}{10}$	Racket	\$95.00		
$\frac{1}{2}$	Tennis ball pack	\$10.00		
Total				

- calculate the number of units required for each item
- find the total dollar value for each
- hence complete the table, calculating the expected total value of all the items.

A picture speaks volumes!

- 5 Guests at Christian and Evol's wedding will each receive an 8.9-cm square picture frame. The frame is designed to fit a photo measuring 4.1 cm by 5 cm at its centre.



- Find the area of:
 - the picture frame
 - the photo
 - the part not covered by the photo.
- A frame and photo have been organised for each of the 140 people invited, with an extra 20 to cover breakages. The cost of the photos is \$2.50 each and the frames will cost \$3.76 each.
 - Christian and Evol received a 15% discount on the cost of the photos because they ordered so many of them. How much did they pay for all the photos?
 - Calculate the cost to Christian and Evol for all the complete frame and photo sets.
 - Hence, find the average cost per gift. Give your answer correct to the nearest 10 cents.

Did you get me a catalogue?

Art Industries is staging an event at which Art Masters are exhibited in four art galleries around the world.

- 6
- Use the Fast Gallery Facts to calculate the potential number of visitors to all 4 galleries over a 12-month period.
 - Each gallery will exhibit the Art Masters show for 3 months. Normally, visitors to a gallery are equally spread throughout the year but it is expected that there will be 20% more visitors when the show is on. Calculate how many people could visit the exhibit across all 4 galleries.
 - The catalogue that complements the exhibit retails for \$40. Printing cost per catalogue, including freight, is \$18. Art Industries calculates that 25% of potential patrons will purchase a catalogue. Calculate the margin that Art Industries are expecting from the sale of all the catalogues.
 - Art Industries are donating 5% of the total sales from catalogues to the World Agricultural Group. Calculate this figure.

... FAST GALLERY FACTS

The number of million visitors per year:

- The French National Gallery: 8.2
- The National Gallery, London: 6.5
- The New Yorker Gallery: 8.6
- The Australian National Gallery of Art: 4.1

A **sales margin** is the difference between the retail price and the cost price.

Many companies are insisting that their carbon footprints be reduced. Being kinder to the planet means not spending so much money on travel! Virtual events give people the chance to use special software to attend an event without having to travel to face-to-face events. They also allow people from all over the world to attend at any time of the day, any day of the week.

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

The Internet - virtually the place to be!

- 1 The Internet is a great place to run an event because so many people use it. How many hours per day on average are these Australians spending on the Internet:
 - a 16- and 17-year-olds?
Idea: $\frac{3.15 \times 1.3 \times 2.9}{3.77}$
 - b 8- to 11-year-olds?
Idea: $\frac{3.15 \times 1.3 \times 2.9}{10.15}$
 - c 12- to 17-year-olds?
Idea: $\frac{3.15 \times 1.3 \times 2.9}{4.55}$
- 2 a What percentage of teenagers in Australia is using social networking sites, such as MySpace and Facebook, to chat with friends from school?
Idea: $\frac{27}{50}$
 - b What percentage of teenagers in Australia is using social networking sites to make new friends?
Idea: $\frac{68}{400}$
- 3 Thirty million people across 150 countries will use a social network to run trade events for selling goods. How many people per country on the average is that?

Web events!

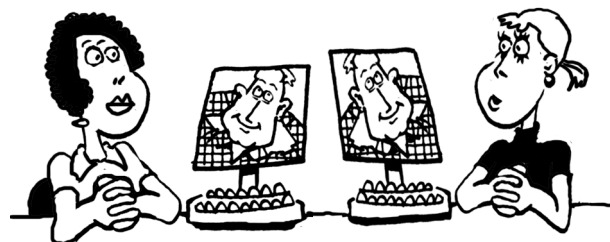
- 4 'Having a web event is greener than having a green event!' says Virtual Events Inc. 'The average avatar, left on all day every day, consumes only 75 kWh of energy per year.' How far would a family car have to travel to use the same energy as an avatar in a year?

Idea: $\frac{2^4 \times 10^6 \times 10^4}{10^2 \times 10^7}$ km

- 5 Jake's publishing company organised a virtual sales event for 7500 people. The website looked just like a real expo. It cost Jake \$35 000 to run the event. How much should he charge each of the people who attend so they he covers his costs and 'breaks even'?
- 6 A fashion event is to be held on the web and Tony has decided to take a booth. The price of a booth depends on its position. The event will be held over 3 virtual floors, with each floor having 18 booths. Some are still available, and Tony looks at the event planner, below.
 - a How many exhibitors can be housed altogether at the web event?
 - b What is the average price of the available booths on this floor?
 - c Each entry 'ticket' is \$25 and 500 people are expected to attend. How much will the event managers receive from the sale of tickets?

	Enter	Enter
1	Virtual booth for sale: \$8000	Virtual booth for sale: \$7000
2	Virtual booth for sale: \$7000	Virtual booth for sale: \$7000
3	Virtual booth for sale: \$6000	Virtual booth for sale: \$6000
4	Virtual booth for sale: \$6000	Virtual booth for sale: \$5000
5	Virtual booth for sale: \$5000	Virtual booth for sale: \$5000
6	Virtual booth for sale: \$5000	Virtual booth for sale: \$5000
	Click here for next floor	Click here for next floor

- 7 Statistics have shown that on the average 70% of people who register for a virtual expo will not attend. Of those who turn up, 50% of visitors will request further information about a product on show. If 5300 people registered for the Extravaganza Expo for exercise equipment, how many would you expect to request further information?



- 8 www.4sickkids.com is a website that allows fundraisers to take payments, accept donations, sell products and organise events. The charge is \$35 per month and 25 cents per transaction (payment or donation) with a set-up fee of \$350. The Kids Foundation uses this website for 18 months, receiving 500 donations averaging \$22.50. Find:



- how much they received in donations
- how much they paid to the website in fees and charges for collecting the donations
- how much was raised for the charity after fees and charges had been paid.

- 9 MegaBit marketing hosted 200 virtual events last year and collected the data in the table, right.
- What percentage of those who paid actually attended? Give your answer correct to 2 decimal places.
 - On average, how many Australians attended the event?
 - How many international people attended the event?
 - What percentage of all the attendees is that?
 - If there were 70 displays, how many minutes on the average did people stay at each display? Give your answer correct to 2 decimal places.

Average number of people who paid to attend the event	3 102
Average number of people who attended the event	1 587
Average time people stayed at the event	2 hours 31 mins
Percentage of Australians at the event	58%

- 10 GNU Events website provided the following information (right) to their customers. Dojo Interiors had 57 people attend their trade show hosted by GNU. If the information GNU provided was correct:

- how many people registered?
- how many people saw the advertising?

GNU Events

For every 1000 people who see your advertising on our web page:

- 20 will click to go to the page to register
- 10 will then register
- 3 will attend.

- 11 Canberra University will organise a web seminar or 'webinar' event next Tuesday at 10.30 am Eastern Standard Time. Minna lives in Denver, Colorado, in the USA. Checking the web, she finds the time differences shown in the Fast Time Facts. What time should she log on to the webinar from Denver?

... FAST TIME FACTS

At 17.02.47 Sunday 30 August 2009 in Canberra, Australia, it is 01.02.47 Sunday 30 August 2009 in Denver, Colorado, USA.



Advertising and promotion are important aspects of event management. The success of an event depends on effective communication to the right people, within the right time frame and with the right frequency.

Tune-in folks!

- 1 Approximately 6.5 million Australians tune in to breakfast radio each day across 5 commercial networks. Of the national figure, 35% of listeners are from New South Wales and 30% from Victoria.
 - a Calculate the current daily listeners across the 5 networks for:
 - i New South Wales
 - ii Victoria.
 - b Breakfast radio airs from 5 am to 9 am, 7 days a week. The average listener tunes in for 2 hours each day. For how many hours per week does the average listener tune in?
 - c The advertising cost during a network’s breakfast show is \$450, which includes two 15-second time slots each hour of the show. Calculate the advertising cost to the promoter for a 10-day advertising plan across 5 networks, including 3.5 hours of voice-over. Voice-over work generally costs \$220 per hour.
 - d How many radio advertisements for the promoter would the average listener be exposed to during the 10-day event?



... FAST ADVERTISING AND PROMOTION FACTS

Advertising and promotion campaigns are made up of two parts:

- the type of advertising media used, such as television, newspapers and radio
- the production of materials used in advertising campaigns, such as television commercials, layout of print design and voice-over work.

The **circulation** is the number of newspapers sold each day.

Depending on the type of event being managed, online advertising can be used.

The pressing issue!

- 2 A Brazilian newspaper called *Il Giornalese* is being used to promote Il Carnivale, held in its capital city. *Il Giornalese*’s advertising rates are shown in Australian dollars in Fast Advertising Facts.
 - a Find the price for a ½-page and a full-page advertisement on Sundays if the costs are in the same proportion as for the other days.
 - b A two-page spread has been bought for Wednesday’s Art section. What would this cost the promoters?
 - c A full page has been bought for Saturday’s paper for three consecutive Saturdays leading up to Il Carnivale. What will this cost the promoters?

... FAST PRESS ADVERTISING FACTS

Day	¼ page	½ page	Full page
Each weekday	\$6 930.00	\$10 395.00	\$15 592.50
Saturdays	\$10 395.00	\$15 592.50	\$23 388.75
Sundays	\$7 485		

- d** The promoters also purchase a $\frac{1}{2}$ page on Page 3 for two consecutive Sundays prior the event. What will all the Sunday advertising cost if Page 3 costs 15% more than the normal price per issue?
- e i** What is the total amount spent by the promoters to advertise in *Il Giornalese*?
- ii** They have forgotten the extra 15% of the total they should have paid for the design of the advertisements. How much will that mean they will spend to promote *Il Carnivale*?
- f** *Il Giornalese* offers a free $\frac{1}{4}$ -page weekday advertisement and a free $\frac{1}{4}$ -page Sunday advertisement. What percentage of the total advertising money spent does this represent? Give your answer correct to 3 decimal places.
- c** Two weeks prior to the event, the promoters endorse an arts program being aired in the evening. The package comprises six 30-second spots and two 15-second spots during the program. They also buy two 15-second prime time ads and one 30-second prime time commercial. How much did they spend?
- d** One week prior to the event, promoters want to target as many potential viewers as possible by purchasing the following spots. How much do they spend on each?
- i** Six 30-second prime time spots
- ii** Eight 15-second prime time spots
- iii** Ten 15-second evening spots
- iv** Twelve 30-second late night spots
- e** Calculate the total amount spent on television advertising.

- f** The combined ticket and merchandise sales for the Art Masters show were very successful nationally. This meant that what the promoters spent on TV advertising was only 4% of the amount returned from the events. What was the total events return?

Plasma? Isn't that blood?

Television is a powerful advertising medium because it potentially has wide audience reach. A 'viewer profile' usually includes where viewers live, what they buy, their ages and their income levels. Advertisers match this profile to the product being promoted and the program being televised.

- 3** The Art Masters fine art show is travelling to all state galleries around Australia. A national advertising campaign covering all states needs to be organised.

- a** One month prior to the event, the promoters purchase two 30-second morning spots, two 30-second evening spots and two 15-second mid-afternoon spots. Use Fast TV Advertising Facts to find out how much they spent.
- b** Three weeks prior to the event, the promoters purchase four 30-second evening spots, one 15-second prime spot, two 30-second mid-morning spots and six 15-second late night spots. How much did they spend?

... FAST TV ADVERTISING ACTS

Advertising rates are based on viewing schedules. Costs vary depending on the show, ratings and demand.

When a promoter buys the advertising rights to claim that a program 'is brought to you by ...', it is called a **program endorsement**.

Time slot	one 15-second spot	two 15-second spots	one 30-second spot	two 30-second spots
Morning: 6 am–9 am	\$1 000	\$1 500	\$1 500	\$2 250
Mid-morning: 9 am–12 pm	\$500	\$650	\$1 000	\$1 300
Midday: 12 pm–3.30 pm	\$500	\$650	\$2 000	\$2 600
Mid-afternoon 3.30 pm – 5 pm	\$3 000	\$4 800	\$5 000	\$8 000
Prime time: 5 pm–7 pm	\$8 000	\$13 600	\$10 000	\$17 000
Evening: 7 pm–11 pm	\$4 000	\$6 600	\$6 000	\$9 900
Late night: after 11 pm	\$350	\$420	\$500	\$600



1 Event fact and fiction

- 1 \$1 513 514 2 140
 3 a 4.8 m² b 833
 4 a 8 b 750 000 or ¾ million c 1932
 5 6 million 6 240 000
 7 300 8 19
 9 1876
 10 a 1647 b 786.384 m c $19\frac{13}{23} \approx 19.57$ ms⁻¹
 11 6.4 km 12 480 13 2266 m²
 14 6.634
 15 a 617 088 b 1607
 16 35.85%; no, one third would be 33.33%
 17 \$109.89

2 Big events, big money

- 1 1 000 000 001 001 000 001 001 000 000
 2 10¹⁰⁰
 3 10^{googol} or 10^{10¹⁰⁰}
 4 \$54 545 455 5 \$21 087
 6 a 2.25 b 11.84% c 325
 7 \$1 185 000
 8 \$300 000 000 000; 0.3 trillion
 9 a 144 000; 144 000; 7 488 000; \$7 488 000
 b \$31 536 000 c 31 years
 10 1818 11 \$510 million and 625 000 staff hours
 12 a \$574.63 b 25.97%
 13 a 525 900 b \$6085

3 Things you need to know to manage events

- 1 12 2 19
 3 a 24 b 32
 4 between 10 and 12 5 6
 6 a 50 b 16
 7 a 13 b 4 hours
 8 a 108 m² b 1.08 m² c 1.8 m² d 3.6 m²
 9 5 10 a 12 b 13 11 $6\frac{1}{4}$; $4\frac{1}{6}$
 12 a $3\frac{1}{2}$ b 150 mL c 5
 13 600 14 190
 15 a 500 g b 36 c 5400

4 Risky business

- 1 \$1250
 2 a 5% b \$3 800 000
 3 a \$3825 b \$1650 c \$5475
 4 a \$1840 b \$122.67
 5 a 60% b \$292.50 6 80%
 7 a 85% b 44.4%; 55.6%
 8 a \$90 000 b \$50 000

5 Launching a book

- 1 212 – 31 = 181 days 2 \$77.50
 3 a i 80% ii 100% b 60%
 4 a \$575 b 25 May 2015
 5 a i \$2040 ii \$2700 iii \$2280 iv \$2340 v \$2400; i or iii
 b \$38.40
 6 3 hours

6 Show me the money!

- 1 28.5 minutes 2 4.75 minutes 3 19
 4 a \$3200 b \$1700
 5 a 60% b 0.6 c 0.042 d 0.284

- 6 a 24 cm³ b 5 c $\frac{1}{9}$ d 2 kg e 880 cm² f 50 cm
 g $7\frac{1}{2}$ times h 700 mL i \$60

- 7 Team Surfer won by 3 points, 8 to 5

7 Jackson's home run

- 1 a 589 b 972 c 7 miles d 11 664 km 2 4
 3 a \$17.50; \$90; \$0; \$1.60; \$0 b \$109.10
 c 11.5 hours
 4 a 972 b 243 litres c \$34 992 d \$34 882.90
 5 a 763 b 9156 c \$27 468 d \$27 358.90
 6 a \$421 b 75 people c \$365 d 121.6 km

8 Farewell to Frances

- 1 a 22 b 9 November c 75 days
 2 \$10 800
 3 a \$30 b \$13 500
 4 a \$25 200 b \$56 c \$15 750 d \$35 e Easy Events
 5 \$14 000–\$20 000
 6 a-b:

Item	Cost	Cost for 320 guests	Cost for 250 guests	Cost for 100 guests
Room hire	\$1000	\$1 000	\$1 000	\$1 000
4-hours of canapés	\$100 per person	\$32 000	\$25 000	\$10 000
Drinks	\$50 per person	\$16 000	\$12 500	\$5 000
1 wait staff	\$35 per hour	\$140	\$140	\$140
DJ for music	\$100 per hour	\$400	\$400	\$400
	Total cost	\$49 540	\$39 040	\$16 540
	Per person	\$154.81	\$156.16	\$165.40

- 7 a 160 m² b 20 c-d:

Item	Cost	Cost for 320 guests	Cost for 250 guests	Cost for 100 guests
Marquee hire	\$1350	\$1 350	\$1 350	\$1 350
'Gold star' finger food (18 items each)	\$27 per person	\$8 640	\$6 750	\$2 700
Drinks package	\$23 per person	\$7 360	\$5 750	\$2 300
Wait staff	\$0 per hour	\$0	\$0	\$0
Music students	\$0 per hour	\$0	\$0	\$0
	Total cost	\$17 350	\$13 850	\$6 350
	Per person	\$54.22	\$55.40	\$63.50

9 Big yachting event

- 1 a 48 people b 72 L c i 51 L ii 412 mL d 200.4 L
2 a 64.795 nm b 18.52 km per hour
c i 6.4795 hours ii 6 hours 29 minutes
3 Yes
4 a 13 hours b 18.03 knots c S33°41'E
5 a 0.27 nm b 48 minutes c 300 m

10 Ready, set, dive!

- 1 a 170 b \$25 500 c \$5 850 d \$31 350 e 5010 km
2 a 3400 kg b 1190 kg c \$195 d Jessica \$39;
Cameron \$53.63
3 a \$44 850 b \$232.88 c \$11 644 d \$379.50
e \$71 674
4 a \$128.25 b \$142.50 c 5 d \$30 e \$150 f \$500
g \$150 h \$192 i \$165 j \$822.75 k \$937
l \$18 329

11 Going to the Tivoli

- 1 a 1.414 b 594.5 mm × 420.5 mm; 1.414
c 297.25 mm; 210.25 mm d $\frac{1}{16}$; 5 grams
e 50%
2 2500 g
3 a A3: 297 mm × 420 mm b i 124 740 mm²
ii 0.12474 m² c 37.42 g
4 a 210 mm × 148.5 mm b cardboard; because it
is more than 200 gsm c thickish d \$3670
e \$4037 f \$4235
5 a Spot Cards 5000 b Premium + 500
c Yes, Artboard d No

12 Bending the ball!

- 1 a 53 000 b 2500 c 20 000 m² d 1000
e 19 000 000
2 a Yes b Yes
3 a 773 b 66 toilets, 275 urinals and 187 washbasins
for men and 374 toilets and 187 washbasins
for women.
4 a 9 pm b 7.49 pm c 55 min
5 a Club Brasserie b Club Brasserie
c Arena Viewing Room d 800
6 \$1 500 000 (20% of gross is \$1 060 000)
7 \$1 419 000 8 \$62 500

13 Opening the racing season

- 1 a 45 mins b i 37½ minutes ii Yes c 4; 15 mins
d 10½ hours
2 a 24 300 cm² b \$220 c \$90.53
3 a \$1668 b \$208.50
4 36 m
5 a 30 m; 36 m b 2.3 km
6 a 720 b $\sim 1.658 \times 10^{34}$

14 Anyone for tennis?

1. a

Section	Ticket sales (\$)
A. Light grey (outer)	\$1 188 000
B. Light grey (inner)	\$168 000
C. Dark grey	\$64 800
D. White (outer)	\$69 000
E. White (inner)	\$116 000

b \$1 605 800

c

Section	% of sales
A. Light grey (outer)	73.98
B. Light grey (inner)	10.46
C. Dark grey	4.04
D. White (outer)	4.30
E. White (inner)	7.22

d \$16 058 000

e i \$2880 ii \$4640 f \$5 700 590

- 2 a i \$ 3654 ii \$1815 b \$5469 c 136 hours
d \$40.21

- 3 a \$21 700 b \$480 000 c \$219 600

15 Fashion Week

- 1 a 280 000 people b \$53 000 000 c 5200 people
d 36 400 people e \$11 778 per fashion buyer

- 2 a \$2500

b-c

Display stand options	Total hire fee	Total floor space
Option A: 1.3 m × 3 m	\$2 000 000	3 900 m ²
Option B: 2.6 m × 3 m	\$1 750 000	3 900 m ²

d \$3 750 000 e 7800 m² f \$6794.87 per m²

- 3 a 5 shows b 35 c i \$24 000 per show
ii \$120 000 a day iii \$840 000

- 4 a \$187 500 b \$370 c \$71.15

- 5 a ~ \$64 000 b ~\$2286 per show c \$18,900
d \$14 400 e 64 pairs f \$16,000 g \$136,300

16 Getting married? Okay, I do!

- 1 a i \$34 125 ii \$18 725 b \$52 250

- 2 a i \$3840 ii \$960 iii \$5170 b i \$720; \$2160
ii \$2910 c i \$1584 ii \$2134 d \$499.50 e \$3150
f \$13 863.50



Female bridal party costs		
a: Bride's outfit		Total:
Base cost	\$3 200.00	\$5 170.00
4 visits to dressmaker @ 5% of base cost	\$640.00	
Veil @ 30% of base cost	\$960.00	
Shoes	\$370.00	
b: Bridesmaids' outfits		Total:
3 dresses @ 22.5% of base cost	\$2 160.00	\$2 910.00
3 pairs of shoes @ \$250	\$750.00	
c: Mother's and mother-in-law's outfits		Total:
2 dresses @ 10% more than bridesmaids' dresses	\$1 584.00	\$2 134.00
2 pairs of shoes @ \$275	\$550.00	
d: Bouquets		
3 bouquets @ 10% of \$185	\$499.50	
e: Gifts		
7 pieces of jewellery @ \$450	\$3 150.00	
f: Total cost for female bridal party	\$13 863.50	

- 3 a \$3500 b \$649.30
c

Product/service	No.	Hire or purchase fees	Total charge
Car hire, per car per day	3	\$235.00	\$705.00
Suit hire, per outfit	7	\$110.00	\$770.00
Photography package	1	\$3 500.00	\$3 500.00
Photographer's assistance (10% of package)	1	\$350.00	\$350.00
Hairdressers, per day	2	\$250.00	\$500.00
Stylist	2	\$275.00	\$550.00
Invitations, including design, printing and postage	140	\$3.10	\$434.00
The band	1	\$1 500.00	\$1 500.00
DJ (30% of band hire fee)	1	\$450.00	\$450.00
Master of Ceremonies	1	\$360.00	\$360.00
Flowers for suit lapels	7	\$8.50	\$59.50
		Total	\$9 178.50

d \$9178.50 e \$13 327.80

- 4 \$79 441.30

17 Modern art!

- 1 a \$2725 b \$173 400 ii 23.875% c i \$8280
ii \$4080 d i \$126 600 ii \$75 960
2 a 1040 invitations b 160 invitations

c i \$49 000 ii \$29 400 d \$8400 e 293 f i 15
ii \$15 750

3 a i \$42 000 ii \$16 800 b \$1486.10 c 8.85%

4 a \$78 225 b \$39 112.50 c Painting: \$10 000;
sculpture: \$20 000; photography: \$5000 d 10.51%

18 All is fair in the streets!

1 a \$420 000 b \$336 000 c \$53 760 d \$55 000
e \$335 240

2 a \$60 000 b \$150 000 c \$96 000 d \$50 000
e \$22 500 f \$21 500

3 a \$3.75 b \$16 957.50 c \$20 187.50 d i 40%
ii \$17 400 e \$37 587.50 f \$76 562.50 g \$38 975

4 a 36 000 people b \$45 000 c i \$208 800
ii \$14 616

5 a \$12 675 b \$5,850

19 Managing event merchandise!

1 \$15 500

2 a \$5280 b \$335 280 c \$11 176

3 a \$36 400 b \$46 800 c \$35 100 d \$22.75

4 a-c

Fraction of people	Product	Retail price	Quantity	Total value
$\frac{1}{4}$	T-shirt	\$28.00	3 000	\$84 000
$\frac{1}{3}$	Cap	\$15.00	4 000	\$60 000
$\frac{1}{10}$	Racket	\$95.00	1 200	\$114 000
$\frac{1}{2}$	Tennis ball pack	\$10.00	6 000	\$60 000
Total				\$318 000

5 a i 79.21 cm² ii 20.5 cm² iii 58.71 cm² b i \$340
ii \$941.60 iii \$5.90

6 a 27.4 million potential visitors
b 8.22 million potential visitors c \$45 210,000
d \$4 110 000

20 A virtual event? Get real!

1 a 3.15 b 1.17 c 2.61

2 a 54% b 17%

3 200 000

4 160 km

5 \$4.67

6 a 54 b \$6000 c \$12 500 7 795

8 a \$11 250 b \$1105 c \$10 145

9 a 51.16% b 920 c i 667 ii 42% d 2.16 minutes

10 a 190 b 19 000 11 Monday 6.30 pm

21 Getting ready to market!

1 a i NSW: 2 275 000 ii Vic: 1 950 000 b 14 hours
c \$23 270 d 40

2 a \$11 227.50; \$16 841.25 b \$31 185 c \$70 166.25
d \$25 823.25 e i \$127 174.50 ii \$146 250.67
f 9.856%

3 a \$16 950 b \$30 360 c \$59 900 d i \$51 000
ii \$54 400 iii \$33 000 iv \$ \$3,600 e \$ 249 210
f \$6 230 250

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A soccer competition – planning and running functions, venue hire
Managing a yachting event – safety, planning, equipment
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An imprint of Blake Education Pty Ltd
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ISBN: 978-1-92571-018-2

